



Department of Public Health and Environment

Air Quality Control Commission

REGULATION NUMBER 30 TOXIC AIR CONTAMINANTS

5 CCR 1001-34

[Editor's Notes follow the text of the rules at the end of this CCR Document.]

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Pursuant to Colorado Revised Statutes section 24-4-103(12.5), any materials incorporated by reference are available for public inspection during normal business hours, or copies may be obtained at a reasonable cost from the Air Quality Control Commission, 4300 Cherry Creek Drive South, Denver, Colorado 80246-1530. Any federal material incorporated by reference is also available through the United States Government Printing Office, online at www.govinfo.gov. Materials incorporated by reference are those editions in existence as of the date indicated and do not include any later amendments.

PART A General Provisions for Toxic Air Contaminants

I. General Provisions

I.A. Definitions

- I.A.1. “Hazardous air pollutant” (HAP) has the meaning provided in section 25-7-103(13), C.R.S.
- I.A.2. “Hazard quotient” (HQ) means a metric to estimate the potential for non-cancer health hazards to occur from exposure to a toxic air contaminant with available Reference Concentrations (RfCs). HQ represents the ratio of an air concentration over the RfC. HQs above 1 mean that the air concentration exceeds the non-cancer RfC.
- I.A.3. “Inhalation unit risk” (IUR) means an estimate of the increased cancer risk from continuous inhalation of a specific toxic air contaminant at a concentration of 1 µg/m³ for a lifetime.
- I.A.4. “Chronic health-protective benchmark” means, for purposes of this Regulation Number 30, a health-based standard, as defined in section 25-7-109.5(1)(e), C.R.S., and informed by the considerations in section 25-7-109.5(6)(b), C.R.S., for each PTAC identified by the Commission, representing an ambient concentration below which adverse health effects are unlikely to occur from exposure only to the individual, relevant PTAC.
- I.A.5. “Priority toxic air contaminant” (PTAC) has the meaning provided in section 25-7-109.5(1)(f), C.R.S.
- I.A.6. “Reference concentration” (RfC) means an estimate (with uncertainty spanning perhaps an order of magnitude) of a continuous inhalation exposure concentration to a specific toxic air contaminant that is likely to be without risk of deleterious effects to people (including sensitive subgroups) during a lifetime of exposure to that individual air contaminant.

PART B Priority Toxic Air Contaminants

- I.A. As of January 17, 2025, the Commission identifies the following five priority toxic air contaminants, as also reflected in Appendix A.
 - I.A.1. Benzene
 - I.A.2. Chromium Compounds, Hexavalent
 - I.A.3. Ethylene Oxide

I.A.4. Formaldehyde

I.A.5. Hydrogen Sulfide

II. Chronic Health-Protective Benchmarks

II.A. The Commission proposes the chronic health-protective benchmarks in Appendix B for each PTAC identified in Appendix A. The chronic health-protective benchmarks will take effect upon General Assembly approval.

III. Emission Control Measures

The following sections establish emission control regulations for each of the priority toxic air contaminants identified in Section I. of this Part B. Each priority toxic air contaminant section contains relevant applicability, exemptions, definitions, emission controls or limitations, recordkeeping, and reporting requirements.

III.A. Benzene

III.A.1. Emission limitations

III.A.1.a. An owner or operator of equipment subject to Regulation Number 24, Part B, Section VI.C. (Petroleum Refinery Equipment Leaks) must comply with the equipment leaks requirements and associated monitoring, recordkeeping, and reporting requirements therein.

III.B. Chromium Compounds, Hexavalent

III.B.1. Applicability

III.B.1.a. An owner or operator of a stationary source with decorative chromium electroplating operations that commences construction before June 14, 2026, and uses or emits hexavalent chromium (chromium compounds, hexavalent).

III.B.1.b. An owner or operator of a stationary source with functional chromium plating operations that uses or emits hexavalent chromium (chromium compounds, hexavalent).

III.B.1.c. An owner or operator of a stationary source with decorative chromium plating operations that commences construction or is modified, where the modification increases use of hexavalent chromium, on or after June 14, 2026.

III.B.2. Exemptions

III.B.2.a. Equipment or activities that are not required to be reported on TAC reports under Regulation Number 3, Part A, Section IX.A.2. or Regulation Number 7, Part B, Section V. are not subject to the requirements in Section III.B.

III.B.3. Definitions

III.B.3.a. “Building enclosure” means a permanent building or physical structure, or portion of a building or physical structure, that is enclosed with a floor, walls, and a ceiling or roof such that airflow is limited from the enclosure to the exterior. This can include the enclosed portion of the building itself or an enclosure within a building that is enclosed such that airflow is limited from the enclosure to the rest of the building. The building enclosure may have limited building enclosure openings to allow access for people, vehicles, equipment, or parts.

III.B.3.b. “Chromic acid anodizing operations” means the electrolytic process by which an oxide layer is produced on the surface of a base material for functional purposes (e.g., corrosion resistance or electrical insulation) using a chromic acid solution. In chromic acid anodizing, the part to be anodized acts as the anode in the electrical circuit, and the chromic acid solution, with a concentration typically ranging from 50 to 100 grams per liter (g/L), serves as the electrolyte.

III.B.3.c. “Chromium plating operations” means the process of performing chrome plating, including: decorative chrome plating, hard chrome plating, and chromic acid anodizing.

III.B.3.d. “Decorative chromium plating operations” means the process by which a thin layer of chromium (typically 0.003 to 2.5 micrometers) is electrodeposited on a Base Material to provide a bright surface with wear and tarnish resistance. In this process, the base material serves as the cathode in the electrolytic cell and the solution serves as the electrolyte. Typical current density applied during this process ranges from 540 to 2,400 amperes per square meter (Amp/m²) for total plating times ranging between 0.5 to 5 minutes.

III.B.3.e. “Functional chrome plating” means hard chromium plating and chromic acid anodizing operations.

III.B.3.f. “Hard chrome plating” means a process by which a thick layer of chromium (typically greater than 1.0

micrometers) is electrodeposited on a base material to provide a surface with functional properties such as wear resistance, a low coefficient of friction, hardness, and corrosion resistance. In this process, the base material serves as the cathodes in the electrolytic cell and the solution serves as the electrolyte. The hard chrome plating process is performed at current densities typically ranging from 1,600 to 6,500 amp/m² for total plating times ranging from 20 minutes to 36 hours depending upon the desired plate thickness. This definition also includes continuous passivation.

III.B.3.g. “High efficiency particulate arrestor (HEPA) filter” means filter(s) rated at 99.97 percent or more efficient in collecting particle sizes 0.3 micrometers.

III.B.3.h. “HEPA vacuum” means a vacuum that is both designed for the use of and fitted with a HEPA filter.

III.B.3.i. “Protected opening method” means any of the methods listed below that restricts air from escaping the building enclosure including a door that automatically closes; overlapping plastic strip curtains that cover the entire opening; vestibule; or airlock system.

III.B.3.j. “Sensitive receptor” means any residence including private homes, condominiums, apartments, or living quarters; education resources such as schools; daycare centers; or health care facilities such as hospitals or retirement or nursing homes. A sensitive receptor includes a long-term care hospital, hospice, prison, or dormitory or similar live-in housing facility.

III.B.3.k. “Stationary source” has the meaning set forth in Regulation Number 3, Part A, Section I.B.52.

III.B.4. Emission limitations and work practices

III.B.4.a. No owner or operator of a stationary source with operations as described in Section III.B.1.c., may use hexavalent chromium for decorative chrome plating operations.

III.B.4.b. Beginning June 14, 2026, or upon commencement of operation if such date is after June 14, 2026, owners or operators of stationary sources with operations as described in Section III.B.1.a, or Section III.B.1.b, that emit hexavalent

chromium must comply with the following housekeeping practices.

III.B.4.b.(i) Chromium acid powder or flakes or other substances that may contain hexavalent chromium must be stored in a closed container in an enclosed storage area.

III.B.4.b.(ii) Chromium acid powder or flakes must be transported from the enclosed storage area to the chrome electroplating or anodizing bath(s) in a closed container.

III.B.4.b.(iii) Any liquid or solid material that may contain hexavalent chromium that is spilled must be cleaned up using one or more of the methods listed in Sections III.B.4.b.(iv)(A) through III.B.4.b.(iv)(E) or contained within one hour after being spilled.

III.B.4.b.(iv) Surfaces within the enclosed storage area, floors and other surfaces in the chrome plating or anodizing tank(s) process area, or any other surface potentially contaminated with hexavalent chromium must be cleaned at least once every seven (7) days with one or more of the following methods. Floors within a 20-foot radius of any buffing, grinding, or polishing workstation(s) must be cleaned at the end of each day on days when buffing, grinding, or polishing are conducted using one or more of the following methods.

III.B.4.b.(iv)(A) HEPA vacuuming.

III.B.4.b.(iv)(B) Hand wiping with a damp cloth.

III.B.4.b.(iv)(C) Wet mopping.

III.B.4.b.(iv)(D) Wet washing.

III.B.4.b.(iv)(E) Low pressure spray nozzle (i.e., water spray nozzle capable of regulating water pressure such that it does not exceed 35 pounds per square inch).

III.B.4.b.(v) Fluid containing hexavalent chromium that adheres to parts when they are removed from the tank must be minimized by adhering to the following best management practices.

- III.B.4.b.(v)(A) For automated lines, drip trays or other containment devices must be installed between chrome plating tanks so that the liquid does not fall through the space between tanks.
- III.B.4.b.(v)(B) For non-automated lines, each part that is dipped in the chrome plating tank for the purposes of chrome plating must be handled so that chromic acid does not drip outside of the tank, unless the liquid is captured by a drip tray or other containment device. Drip trays or containment devices must capture and return the liquid to the tanks and be cleaned weekly such that there is no accumulation of visible dust or residue on the drip tray or other containment device.
- III.B.4.b.(vi) Buffing, grinding, or polishing areas:
- III.B.4.b.(vi)(A) Buffing, grinding, or polishing areas must be separated from the electroplating or anodizing operation by a physical barrier (e.g., plastic strip curtains) that restricts air flow out of these areas.
- III.B.4.b.(vi)(B) Beginning on or after June 14, 2028, or upon commencement of operation if such date is after June 14, 2026, buffing, grinding, or polishing areas shall be conducted within a building enclosure that meets the requirements in Section III.B.4.c. for as long as the facility continues to use hexavalent chromium.
- III.B.4.b.(vii) Chromium or chromium-containing wastes must be stored, disposed of, recovered, or recycled using practices that do not lead to fugitive emissions. Containers with chromium-containing waste must be kept closed at all times, except when being filled or emptied, and stored in an enclosed storage area.
- III.B.4.b.(viii) Beginning December 14, 2026, parts being chrome plated or equipment that was previously in a chrome plating tank cannot be spray rinsed unless the parts or equipment are fully lowered inside a tank such that the liquid from the spray rinsing is captured inside the tank. If the part cannot be fully lowered

inside the tank for spray rinsing, splash guards made of non-permeable and durable material (e.g., metal, plastic) must be installed at the tank to minimize overspray and ensure that any hexavalent chromium laden liquid is captured and returned to the tank where parts are sprayed to remove excess chromic acid to ensure that any hexavalent chromium laden liquid is returned to the tank. Splash guards must be cleaned weekly.

III.B.4.b.(ix) Beginning December 14, 2026, compressed air cleaning or drying operations cannot be performed within 15 feet of any chrome plating tank(s), unless a barrier separates the compressed air cleaning or drying operation from the tank(s). A tank wall may function as the barrier provided the parts being air cleaned or dried are below the lip of the tank.

III.B.4.c. Beginning June 14, 2028, or upon commencement of operation if such date is after June 14, 2026, owners or operators of stationary sources with operations described in Section III.B.1.a., or Section III.B.1.b. that emit hexavalent chromium must comply with, at a minimum, the following building enclosure requirements.

III.B.4.c.(i) All hexavalent chromium plating or anodizing tanks must be operated only within a building enclosure.

III.B.4.c.(ii) All building enclosure openings that are open to the exterior and on opposite ends of the building enclosure from each other must be equipped with a protected opening method and cannot be simultaneously open except during the passage of vehicles, equipment, or people through the building enclosure opening.

III.B.4.c.(iii) All building enclosure openings that directly face any sensitive receptor that is located within 1,000 feet, as measured from the property line of the sensitive receptor to the building enclosure opening, must be equipped with a protected opening method and remain closed except during the passage of vehicles, equipment, or people through the building enclosure opening.

III.B.4.d. Beginning December 14, 2026, or upon commencement of operation if such date is after June 14, 2026, owners or operators of stationary sources with operations described in Sections III.B.1.a. or III.B.1.b., must use the following list of approved chemical fume suppressants and adhere to the associated surface tension limits of such approved chemical fume suppressant.

Table 1. Approved Chemical Fume Suppressants and Surface Tensions			
Chemical Fume Suppressant and Manufacturer	Application	Stalagmometer Measured Surface Tension (dynes/centimeter)	Tensiometer Measured Surface Tension (dynes/centimeter)
Fumetrol 21 LF2® Atotech, U.S.A	Hard plating	≤ 30	≤ 27
Dicolloy CRPF® ProCom LLC	Decorative plating and chromic acid anodizing	≤ 32	≤ 29
HCA-8.4® Hunter Chemical LLC	Decorative plating and chromic acid anodizing	≤ 25	≤ 22
HCA-8.4® Hunter Chemical LLC	Hard plating	≤ 33	≤ 30
Macuplex STR NPFx® MacDermid Enthone Industrial Solutions	Decorative plating and chromic acid anodizing	≤ 32	≤ 30

III.B.4.d.(i) Additional chemical fume suppressants not listed in Table 1 that have been approved by the California Air Resources Board (CARB) under the Innovative Products provisions in division 3, chapter 1, subchapter 7.5, Section 93102.8 (Jan 1, 2024) of Title 17 of the California Code of Regulations are approved for use, provided the suppressants are used in an application that is consistent with CARB's approval and that the stationary source maintains the surface tensions demonstrated as part of that approval.

III.B.5. Performance testing

III.B.5.a. Beginning April 15, 2027, or upon commencement of operation if such date is after June 14, 2026, owners or operators of stationary sources with operations described in Sections III.B.1.a. or III.B.1.b., that emit hexavalent chromium must complete a performance test, according to the following requirements.

III.B.5.a.(i) Owners or operators must follow the performance testing requirements in 40 CFR 63.344 (February 27, 2014).

III.B.6. Recordkeeping. For stationary sources with operations described in Sections III.B.1.a. or III.B.1.b, the following records must be kept for a period of five (5) years and made available to the Division upon request.

III.B.6.a. Records demonstrating compliance with the best management practices in Sections III.B.4.b. through III.B.4.d.

III.B.6.b. Records demonstrating compliance with surface tension limits in Table 1 or in CARB's approval under Section III.B.4.d.

III.B.6.c. Records of any submitted reports in Section III.B.7.

III.B.7. Reporting. The following reporting must be submitted in a Division-approved format.

III.B.7.a. Submit the results of the performance test according to the procedures in 40 CFR 63.347 (February 27, 2014).

III.B.7.b. Beginning on June 30, 2028, and annually thereafter, facilities identified in Sections III.B.1.a. and III.B.1.b. of this

Part B that operated in the prior calendar year must submit an annual compliance report that covers the previous calendar year. The report must include the following:

- III.B.7.b.(i) Company information: Include the facility name, address, owner or operator name, and the measured distance to the property boundary of the nearest sensitive receptor. Measurements must be rounded to the nearest foot. For facilities without an add-on air pollution control device, measure from the edge of the chrome plating tank nearest to the receptor. For facilities with an add-on air pollution control device, measure from the centroid of the stack.
- III.B.7.b.(ii) Any deviations to the specified surface tension limits listed either in Table 1 of Section III.B. or the surface tension limits for an approved alternative fume suppressant, including the date of the occurrence, the duration of the occurrence, the cause of the occurrence, the amount of excess emissions, and a copy of the record(s) where the deviation was recorded.
- III.B.7.b.(iii) A certification verifying the facility has followed all requirements applicable to their source as listed in III.B. If the requirements were not followed, include an explanation of the reasons for the deviations, and a copy of any applicable records associated with the deviation.
- III.B.7.b.(iv) A certification that the procedures in the facility operations and maintenance plan prepared under 40 CFR 63.342(f)(3) (Sept. 19, 2012) were followed.

III.C. Ethylene Oxide

III.C.1. Applicability

- III.C.1.a. A stationary source that is a sterilization facility that uses or emits ethylene oxide, that commences operation or modification before, on or after July 1, 2026.

III.C.2. Exemptions

- III.C.2.a. A source is exempt from requirements of Section III.C., provided that the source's total usage of ethylene

oxide is 400 pounds or less during any consecutive 12-month period.

- III.C.2.b. Equipment or activities that are not required to be reported on TAC reports under Regulation Number 3, Part A, Section IX.A.2. or Regulation Number 7, Part B, Section V. are not subject to the requirements in Section III.C.

III.C.3. Definitions

- III.C.3.a. “Sterilization chamber vent” has the same meaning as the term defined at 40 C.F.R. Part 63, Subpart O (April 5, 2024).
- III.C.3.b. “Aeration room vent” has the same meaning as the term defined at 40 C.F.R. Part 63, Subpart O (April 5, 2024).
- III.C.3.c. “Chamber exhaust vent” has the same meaning as the term defined at 40 C.F.R. Part 63, Subpart O (April 5, 2024).
- III.C.3.d. “Group 1 room air emissions” has the same meaning as the term defined at 40 C.F.R. Part 63, Subpart O (April 5, 2024).
- III.C.3.e. “Group 2 room air emissions” has the same meaning as the term defined at 40 C.F.R. Part 63, Subpart O (April 5, 2024).
- III.C.3.f. “Permanent total enclosure” has the same meaning as the term defined at 40 C.F.R. Part 63, Subpart O (April 5, 2024).
- III.C.3.g. “Sterilization facility” has the same meaning as the term defined at 40 C.F.R. Part 63, Subpart O (April 5, 2024).
- III.C.3.h. “Catalytic oxidizer” has the same meaning as the term defined at 40 C.F.R. Part 63, Subpart O (April 5, 2024).
- III.C.3.i. “Stationary source” has the meaning set forth in Regulation Number 3, Part A, Section I.B.52.

III.C.4. Operational standard

- III.C.4.a. Beginning July 1, 2026, the owner or operator of the LivaNova sterilization facility (AIRS ID 059-1258) must limit ethylene oxide usage to 12 tons per year or less in lieu of

complying with Sections III.C.5.a., III.C.6.a., and III.C.7.a. Usage shall be determined on a 12-month rolling average.

This limit shall remain in effect until the source demonstrates compliance with the emission requirements specified in Section III.C.5.a.

III.C.5. Emission limitations

III.C.5.a. An owner or operator of a sterilization facility, as defined in Section III.C.1.a., must comply with Subpart O National Emission Standards for Hazardous Air Pollutants for Ethylene Oxide Sterilization and Fumigation Operations, 40 C.F.R. Part 63, Subpart O (April 5, 2024), unless otherwise specified in Section III.C.5.b.

III.C.5.b. The owner or operator of the LivaNova sterilization facility (AIRS ID 059-1258) must:

III.C.5.b.(i) The catalytic oxidizer which controls emissions from the sterilization chamber vents, aeration room vents, and chamber exhaust vents (i.e., AIRS Point 004 of Permit Number 99JE0094) must meet a 99.5% reduction of ethylene oxide prior to April 6, 2028. The owner or operator must monitor compliance with this condition through the results of approved compliance tests (when required), compliance with the Division-approved operations and maintenance plan, compliance records, and other methods as approved by the Division.

III.C.5.b.(ii) Beginning April 6, 2028, the catalytic oxidizer which controls emissions from the sterilization chamber vents, aeration room vents, and chamber exhaust vents emissions must demonstrate compliance with the applicable standard under 40 C.F.R. Part 63, Subpart O (April 5, 2024) section 63.362(b), (c), (d), (e), or section 63.362(i).

III.C.6. Recordkeeping

III.C.6.a. An owner or operator of a sterilization facility as defined in Section III.C.1.a., must comply with recordkeeping requirements in §63.3687 in Subpart O of the National Emission Standards for Hazardous Air Pollutants for Ethylene Oxide Sterilization and Fumigation Operations, 40 C.F.R. Part 63, Subpart O (April 5, 2024), unless otherwise specified in Section III.C.6.b.

III.C.6.b. The owner or operator of the LivaNova sterilization facility (AIRS ID 059-1258) must:

III.C.6.b.(i) Prior to April 6, 2028, the stationary source must maintain records to demonstrate compliance with the emission limitations in Section III.C.5.b.i., as necessary.

III.C.6.b.(ii) Beginning April 6, 2028, the stationary source must follow the applicable recordkeeping requirements in 40 C.F.R. Part 63, Subpart O (April 5, 2024) pertaining to sterilization chamber vents, aeration room vents, and chamber exhaust vent emissions.

III.C.6.b.(iii) Beginning July 1, 2026, and until April 6, 2028, calculate compliance with applicable yearly operational standards in Section III.C.5.a. on a rolling twelve (12) month total. By the end of each month, the source must calculate a new twelve-month total based on the previous twelve months usage data. The source must record these monthly usage rates and keep a compliance record on site, or at a local field office with site responsibility, for Division review.

III.C.7. Reporting

III.C.7.a. Owners or operators of applicable sterilization facilities, as described in Section III.C.1.a., must comply with reporting requirements in §63.3666 of 40 C.F.R Part 63 Subpart O of the National Emission Standards for Hazardous Air Pollutants for Ethylene Oxide Sterilization and Fumigation Operations, 40 C.F.R. Part 63, Subpart O (April 5, 2024), unless otherwise specified in Section III.C.7.b.

III.C.7.b. The owner or operator of the LivaNova sterilization facility (AIRS ID 059-1258) must:

III.C.7.b.(i) Until April 6, 2028, the stationary source must submit all reports to the Division as necessary to demonstrate compliance with the emission limitations in Section III.C.5.b.(i).

III.C.7.b.(ii) Beginning April 6, 2028, the owner or operator must comply with the reporting requirements in 40 C.F.R. § 63.366(b)(1), (5), and (10) and § 63.366(c) (April 5, 2024) for reporting deviations. The first quarterly compliance report must be submitted to the

Division by July 30, 2028, and must cover the period from April 6, 2028, through June 30, 2028. Thereafter, quarterly compliance reports as required under 40 C.F.R. § 63.366(b)(10) and § 63.366(c) (April 5, 2024) must be submitted to the Division, as provided in 40 C.F.R. Part 63, Subpart O (April 5, 2024) for sterilization chamber vents, aeration room vents, and chamber exhaust vents emissions.

III.D. Formaldehyde

III.D.1. Applicability

Stationary sources with spark ignition stationary reciprocating internal combustion engines or stationary combustion turbines that meet the applicability criteria in Section III.D.1. must comply with relevant emission control requirements in this Section III.D.

III.D.1.a. An owner or operator of a stationary source that commenced construction before September 1, 2026, with stationary source-wide actual emissions of formaldehyde at or above four thousand (4,000) pounds per year (lb/yr), that operates a spark ignition stationary reciprocating internal combustion engine or a stationary combustion turbine, and that is located as described in either Section III.D.1.a.(i) or III.D.1.a.(ii).

III.D.1.a.(i) The stationary source is located in or within 1-mile of a census block that meets the 2020 PTAC Cancer Risk Screening Area; or

III.D.1.a.(ii) The stationary source is located in or within one (1) mile of an occupied area, where the occupied area is, as of September 1, 2026, within the boundary of a disproportionately impacted community.

III.D.1.a.(ii)(A) Alternatively, an owner or operator may assume that the stationary source's location in or within one (1) mile of a disproportionately impacted community as of September 1, 2026, means the stationary source is also located in or within one (1) mile of an occupied area.

III.D.1.a.(iii) The stationary source's actual emissions of formaldehyde are determined as follows

- III.D.1.a.(iii)(A) The highest stationary source-wide actual emissions of formaldehyde recorded on the stationary source's 2023, 2024, or 2025 TAC report (as available). If the stationary source did not submit a 2023, 2024, or 2025 TAC report, the emissions are based on the most recent Air Pollutant Emission Notices (APENs), excluding any cancelled APEN for points that were not operating after June 30, 2026, data (if available), unless an emissions update report is received by January 1, 2027.
- III.D.1.a.(iii)(B) A stationary source may choose to determine actual emissions of formaldehyde based on the three year average of the actual emissions of formaldehyde recorded on the stationary source's TAC reports for calendar years 2023, 2024, and 2025.
- III.D.1.a.(iii)(C) A stationary source that commenced construction prior to September 1, 2026, but did not commence operation as of September 1, 2026, must determine actual emissions of formaldehyde based on the first 12 months after commencing operation.
- III.D.1.b. An owner or operator of a stationary source that commences construction on or after September 1, 2026, that will operate a spark ignition stationary reciprocating internal combustion engine or a stationary combustion turbine, with forecasted actual formaldehyde emissions must submit the forecasted formaldehyde emissions as specified in Sections III.D.1.b.(i) through III.D.1.b.(iii).
- III.D.1.b.(i) Forecasted actual formaldehyde emissions must be submitted in a Division-approved format with the permit application.
- III.D.1.b.(ii) If no permit application is required, the forecasted actual formaldehyde emissions and certification of location assessment must be submitted to the Division in a Division-approved format before the source commences operation of the spark ignition stationary reciprocating internal combustion engine or stationary combustion turbine.

III.D.1.b.(iii) If the forecasted actual formaldehyde emissions meet or exceed four thousand (4,000) pounds per year (lb/yr), the owner or operator of such stationary source must comply with relevant emission control requirements in this Section III.D. upon commencement of operation of the spark ignition stationary reciprocating internal combustion engine or a stationary combustion turbine.

III.D.1.c. An owner or operator of a stationary source that commences construction before September 1, 2026, that undergoes a modification of a spark ignition stationary reciprocating internal combustion engine or a stationary combustion turbine after September 1, 2026, with forecasted actual formaldehyde emissions and that is located as described in either Section III.D.1.c.(i) or III.D.1.c.(ii) must submit the forecasted formaldehyde emissions and location certification as specified in Sections III.D.1.c.(iii) through III.D.1.c.(v).

III.D.1.c.(i) The stationary source is located in or within 1-mile of a census block that meets the 2020 PTAC Cancer Risk Screening Area, as of permit application date or, if a permit application is not required, prior to commencement of construction of the modified source; or

III.D.1.c.(ii) The stationary source is located in or within one (1) mile of an occupied area within the boundary of a disproportionately impacted community as of permit application date or, if a permit application is not required, prior to commencement of construction of the modified source.

III.D.1.c.(ii)(A) Alternatively, an owner or operator may assume that the stationary source's location in or within one (1) mile of a disproportionately impacted community as of permit application date or, if a permit application is not required, prior to commencement of construction of the modified source, means the stationary source is also located in or within one (1) mile of an occupied area.

III.D.1.c.(iii) The owner or operator of the stationary source must conduct the location assessment within thirty

(30) days prior to submitting a permit application for the modification. The owner or operator must certify completion of the location assessment with the permit application in a Division-approved format.

III.D.1.c.(iv) Forecasted actual formaldehyde emissions must be submitted in a Division-approved format with any permit application as required by Regulation Number 3, Part B, that is associated with the modification.

III.D.1.c.(iv)(A) If no permit application is required, the forecasted actual formaldehyde emissions and certification of location assessment must be submitted to the Division in a Division-approved format before the source commences operation of the spark ignition stationary reciprocating internal combustion engine or stationary combustion turbine.

III.D.1.c.(iv)(B) If the forecasted actual formaldehyde emissions meet or exceed four thousand (4,000) pounds per year (lb/yr), the owner or operator of such stationary source must comply with relevant emission control requirements in this Section III.D. upon commencement of operation of the spark ignition stationary reciprocating internal combustion engine or a stationary combustion turbine.

III.D.1.d. Emissions from equipment or activities that are not required to be reported on TAC reports under Regulation Number 3, Part A, Section IX.A.2. (or for ONGAEIR sources: activities, emissions, or equipment not required to be reported under Regulation Number 7, Part B, Section V.), will not be considered in determining whether the stationary source-wide emissions meet or exceed the formaldehyde emission threshold in this Section III.D.1.

III.D.1.e. Definitions, for purposes of determining applicability.

III.D.1.e.(i) "2020 PTAC Cancer Risk Screening Area" means a census block that has a combined modeled cancer risk from carcinogenic priority air toxics at or above 20 in a million excess cancer cases at the time

Regulation 30, Part B, Section III. was adopted and is identified in Division guidance for identifying the 2020 PTAC Cancer Risk Screening Areas (April 2026).

- III.D.1.e.(ii) “Actual emissions” (for the purposes of this Part B, Section III.), has the same meaning set forth in Regulation Number 3, Part A, Section IX.
- III.D.1.e.(iii) “Commenced construction” for a stationary source has the same meaning set forth in Regulation Number 3, Part A, Section I.B.14.
- III.D.1.e.(iv) “Disproportionately impacted community” is identified as any census block group (or in the case of “Mobile home communities” a specific location) identified in the Disproportionately Impacted Community Map (November 2024) after selecting the criteria for “Mobile home communities”, “low-income population above 40%”, “People of color population above 40%”, “Housing cost-burdened population above 50%”, “Linguistically isolated population above 20%”, and “Colorado EnviroScreen percentile score above 80”.
- III.D.1.e.(v) “Forecasted actual formaldehyde emissions” means the highest actual emissions, in tons per year, at which a stationary source described in Sections III.D.1.b. or III.D.1.c. is projected to emit formaldehyde in any one of the five calendar years (twelve-month periods) including and following the commencement of operation of the new or modified stationary source. In determining the forecasted actual formaldehyde emissions, before commencing construction, the owner or operator of the stationary source must consider all relevant information, including but not limited to, historical operational data, the company's own representations, the company's expected business activity and the company's highest projections of business activity, the company's filings with the State or Federal regulatory authorities, and compliance plans.
- III.D.1.e.(vi) “Modified” or “modification” has the same meaning set forth in Regulation Number 3, Part A, Section I.B.33.

III.D.1.e.(vii) "Occupied Area" means (1) a building or structure designed for use as a place of residency by a person, a family, or families. The term includes manufactured, mobile, and modular homes, except to the extent that any such manufactured, mobile, or modular home is intended for temporary occupancy or for business purposes; (2) indoor or outdoor spaces associated with a school that students use commonly as part of their curriculum or extracurricular activities; (3) five thousand (5,000) or more square feet of building floor area in commercial facilities that are operating and normally occupied during working hours; and (4) an outdoor venue or recreation area, such as a playground, permanent sports field, amphitheater, or other similar place of outdoor public assembly.

III.D.1.e.(viii) "Stationary source" has the meaning set forth in Regulation Number 3, Part A, Section I.B.52.

III.D.1.f. Recordkeeping, for the purposes of determining applicability.

The owner or operator of each stationary source as described in Sections III.D.1. must maintain the following records, as applicable, for a period of five (5) years and make records available to the Division upon request.

III.D.1.f.(i) No later than January 1, 2027, an owner or operator of a stationary source that is determining applicability in accordance with Section III.D.1.a. must maintain the following records.

III.D.1.f.(i)(A) Records of the stationary source's actual emissions of formaldehyde and the supporting methodology used to determine emissions in accordance with Section III.D.1.a.(iii), and the date of the initial emissions threshold determination.

III.D.1.f.(i)(B) An owner or operator that is determining stationary source location for the purposes of applicability in Section III.D.1.a. must maintain the following records of location determination:

III.D.1.f.(i)(B)(1) Verification that the stationary source is located in or within

one (1) mile of a 2020 PTAC Cancer Risk Screening Area as of September 1, 2026; or

III.D.1.f.(i)(B)(2) Verification that the stationary source is located in or within one (1) mile of an occupied area within a disproportionately impacted community as of September 1, 2026.

III.D.1.f.(ii) No later than 60 days after the first 12 months of operation, an owner or operator of a stationary source that commenced construction prior to September 1, 2026, but did not commence operation as of September 1, 2026, determining stationary source applicability in accordance with Section III.D.1.a.(iii)(C) must maintain records of stationary source-wide actual emissions of formaldehyde.

III.D.1.f.(iii) An owner or operator that is determining applicability in accordance with Section III.D.1.b. must maintain records of stationary source-wide forecasted actual formaldehyde emissions and the date of the initial applicability emission threshold determination.

III.D.1.f.(iii)(A) An owner or operator that is determining applicability in accordance with Section III.D.1.b. and its forecasted actual formaldehyde emissions are below the 4,000 lb/yr applicability threshold must maintain records of the stationary source-wide annual actual emissions of formaldehyde for each of the five (5) calendar years after commencement of operation of the new stationary source.

III.D.1.f.(iv) An owner or operator that is determining applicability in accordance with Section III.D.1.c., must maintain records of stationary source-wide forecasted actual formaldehyde emissions and the date of the initial applicability emission threshold determination.

III.D.1.f.(iv)(A) An owner or operator that is determining applicability in accordance with Section III.D.1.c. and its forecasted actual formaldehyde emissions are below the 4,000

lb/yr applicability threshold must maintain records of the stationary source-wide annual actual emissions of formaldehyde for each of the five (5) calendar years after commencement of operation of the modified stationary source.

III.D.1.f.(v) Records of reports related to applicability submitted pursuant to Section III.D.1.h.

III.D.1.g. Reports related to applicability.

III.D.1.g.(i) Emission update report: No later than January 1, 2027, if an owner or operator assessing stationary source-wide emissions in accordance with Section III.D.1.a. elects to use the emission update mechanism in Section III.D.1.a.(iii)(A) and stationary source-wide actual emissions of formaldehyde are below 4,000 pounds per year (lb/yr), the owner or operator must submit an update to previously reported emissions in a Division-approved format.

III.D.1.g.(i)(A) The emissions update must verify that the stationary source-wide actual emissions are below the formaldehyde emissions threshold of 4,000 pounds per year (lb/yr) based on calendar year 2025 actual emissions, or based on the three year average of the actual emissions of formaldehyde recorded on the stationary source's TAC reports for calendar years 2023, 2024, and 2025, in accordance with procedures and exemptions set forth in Regulation Number 3, Part A, Section IX., or Regulation Number 7, Part B, Section V.

III.D.1.g.(ii) Notice of Applicability for an existing stationary source that had not yet commenced operation as of September 1, 2026: No later than 60 days after the first 12 months of operation, for an owner or operator of a stationary source that commenced construction prior to September 1, 2026, but had not yet commenced operation as of that date, and the stationary source-wide actual emissions of formaldehyde meet or exceed 4,000 pounds per year (lb/yr), the owner or operator must:

- III.D.1.g.(ii)(A) Submit a notification of applicability to the Division which verifies the calendar year of actual emissions of formaldehyde from the first 12 months after commencing operation that met or exceeded 4,000 pounds per year (lb/yr); and
- III.D.1.g.(ii)(B) Comply with any applicable requirements in this Section III.D. within one hundred and eighty (180) days of the date the notification was due.
- III.D.1.g.(iii) Notice of Applicability for a new stationary source: If a stationary source's forecasted actual formaldehyde emissions are below 4,000 pounds per year (lb/yr), and the source subsequently has actual emissions of formaldehyde at or above 4,000 lb/yr in any of the first five (5) calendar years after operation, no later than March 1 of the subsequent calendar year where actual emissions of formaldehyde meet or exceed 4,000 pounds per year (lb/yr), the owner or operator must submit a notification of applicability in a Division-approved format which verifies the calendar year of actual emissions of formaldehyde that met or exceeded 4,000 pounds per year (lb/yr).
- III.D.1.g.(iii)(A) The owner or operator of such stationary source must comply with any applicable requirements in this Section III.D. within one hundred and eighty (180) days of the date the notification was due.
- III.D.1. g.(iv) Notice of Applicability after a stationary source modifies: If a stationary source's forecasted actual formaldehyde emissions are below 4,000 pounds per year (lb/yr), and the source subsequently has actual emissions of formaldehyde at or above 4,000 lb/yr in any of the first five (5) calendar years after operation, no later than March 1 of the subsequent calendar year where actual emissions of formaldehyde meet or exceed 4,000 pounds per year (lb/yr), the owner or operator must:
- III.D.1.g.(iv)(A) Submit a notification of applicability to the Division which verifies the calendar year of actual emissions of

formaldehyde that met or exceeded 4,000 pounds per year (lb/yr); and

III.D.1.g.(iv)(B) Comply with any applicable requirements in this Section III.D. within one hundred and eighty (180) days of the date the notification was due.

III.D.2. Requirements for Stationary Spark Ignition Internal Combustion Engines.

III.D.2.a. Exemptions.

III.D.2.a.(i) Stationary compression ignition reciprocating internal combustion engines.

III.D.2.a.(ii) Non-road engines, as defined in Regulation Number 3, Part A, Section I.B.

III.D.2.a.(iii) Emergency power generators exempt from APEN or construction permit requirements pursuant to Regulation Number 3, Parts A or B.

III.D.2.a.(iv) Stationary natural gas-fired reciprocating internal combustion engines that commenced construction before September 1, 2026, if the owner or operator will remove or electrify the engine as an emissions reduction measure to comply with Regulation Number 7, Part B, Section VII.

III.D.2.a.(v) Equipment or activities that are not required to be reported on TAC reports under Regulation Number 3, Part A, Section IX.A.2. or Regulation Number 7, Part B, Section V. are not subject to the requirements in Section III.D.2.c. through III.D.2.g.

III.D.2.b. Definitions.

III.D.2.b.(i) "Area HAP source" means a stationary source of HAP that is not a major HAP source.

III.D.2.b.(ii) "Commenced construction" means

III.D.2.b.(ii)(A) For a spark ignition stationary reciprocating internal combustion engine subject to 40 CFR Part 63, Subpart ZZZZ as specified in Sections 63.6585 and 63.6590 (August 30, 2024), the owner or operator has

undertaken or has entered into a contractual obligation to undertake and complete a continuous program of construction. Construction meaning the on-site fabrication, erection, or installation of the stationary internal combustion engine. Construction does not include removal of an engine from an existing location and reinstallation of such engine at a new location.

III.D.2.b.(ii)(B) For a stationary reciprocating internal combustion engine subject to 40 CFR Part 60, Subpart JJJJ (August 30, 2024) as specified in 40 CFR Part 63, Section 63.6590(c) (August 30, 2024), the date the engine is ordered by the owner or operator.

III.D.2.b.(iii) “Fixed capital cost” means the capital needed to provide all the depreciable components of an existing stationary reciprocating internal combustion engine.

III.D.2.b.(iv) “Four-stroke engine” means any type of engine which completes the power cycle in two crankshaft revolutions, with intake and compression strokes in the first revolution and power and exhaust strokes in the second revolution.

III.D.2.b.(v) “Lean burn engine” means any two-stroke or four-stroke spark ignited engine that does not meet the definition of a rich burn engine.

III.D.2.b.(vi) “Major HAP source” has the same meaning as defined in Regulation Number 3, Part A, Section I.B.

III.D.2.b.(vii) “Reciprocating internal combustion engine” (RICE) means any reciprocating internal combustion engine which uses reciprocating motion to convert heat energy into mechanical work and which is not used to propel a motor vehicle or a vehicle used solely for competition.

III.D.2.b.(viii) “Reconstruction” means the replacement of components of an affected or previously non-affected stationary reciprocating internal combustion engine to such extent that (1) the fixed capital cost of the new components exceeds fifty (50) percent of the fixed

capital cost that would be required to construct a comparable new source and (2) it is technologically and economically feasible for the reconstructed source to meet the requirements in this Section III.D.2.

III.D.2.b.(ix) “Rich burn engine” means any four-stroke spark ignited engine where the manufacturer's recommended operating air/fuel ratio divided by the stoichiometric air/fuel ratio at full load conditions is less than or equal to 1.1. Engines originally manufactured as rich burn engines, but modified prior to December 19, 2002, with passive emission control technology for NO_x (such as pre-combustion chambers) will be considered lean burn engines. Also, existing engines where there are no manufacturer's recommendations regarding air/fuel ratio will be considered a rich burn engine if the excess oxygen content of the exhaust at full load conditions is less than or equal to 2 percent.

III.D.2.b.(x) “Spark ignition RICE” means relating to either a gasoline-fueled engine or any other type of engine with a spark plug (or other sparking device) and with operating characteristics significantly similar to the theoretical Otto combustion cycle. Spark ignition engines usually use a throttle to regulate intake air flow to control power during normal operation.

III.D.2.b.(xi) “Startup” means the time from initial start until applied load and engine and associated equipment reaches steady state or normal operation. For stationary engines with catalytic controls, engine startup means the time from initial start until applied load and engine and associated equipment, including the catalyst, reaches steady state or normal operation.

III.D.2.b.(xii) “Stationary reciprocating internal combustion engine” means any reciprocating internal combustion engine which uses reciprocating motion to convert heat energy into mechanical work and which is not mobile. Stationary RICE differ from mobile RICE in that a stationary RICE is not a non-road engine as defined at 40 CFR 1068.30 (April 18, 2024), and is not used to propel a motor vehicle or a vehicle used solely for competition.

III.D.2.b.(xiii) “Two-stroke engine” means a type of engine which completes the power cycle in single crankshaft revolution by combining the intake and compression operations into one stroke and the power and exhaust operations into a second stroke. This system requires auxiliary scavenging and inherently runs lean of stoichiometric.

III.D.2.c. Emissions limitations.

III.D.2.c.(i) Beginning January 1, 2027, unless implementation date is otherwise specified in Section III.D.1.g., the owner or operator of a spark ignition stationary reciprocating internal combustion engine that commenced construction or reconstruction prior to September 1, 2026, at a stationary source described in Section III.D.1.a. must comply with the applicable formaldehyde or carbon monoxide emissions limitations, operating limitations, and associated compliance provisions in 40 CFR Part 63, Subpart ZZZZ, Sections 63.6600 through 63.6603, 63.6635, and 63.6640 (August 30, 2024) or 40 CFR Part 60, Subpart JJJJ, Sections 60.4233 through 60.4235 and 60.4243 (August 30, 2024) that correspond to the engine.

III.D.2.c.(ii) The owner or operator of an engine described in Sections III.D.2.c.(ii)(A) or III.D.2.c.(ii)(B) and with a horsepower equal to or greater than 100 must comply with a carbon monoxide emission limit of 0.40 g/hp-hr, except as provided in Table 1 or Section III.D.2.c.(iii), and operating limitations and associated compliance provisions in 40 CFR Part 63, Subpart ZZZZ, Sections 63.6600 through 63.6603, 63.6635, and 63.6640 (August 30, 2024) or 40 CFR Part 60, Subpart JJJJ, Section 60.4243 (August 30, 2024) that correspond to the engine.

III.D.2.c.(ii)(A) A spark ignition stationary reciprocating internal combustion engine that commences construction or reconstruction on or after September 1, 2026, at a stationary source described in Section III.D.1.a. or a stationary source described in Section III.D.1.c.(iv)(B) that meets the criteria set forth in either Section III.D.1.c.(i) or III.D.1.c.(ii).

III.D.2.c.(ii)(B) A spark ignition stationary reciprocating internal combustion engine that commences construction at a stationary source described in Section III.D.1.b.(iii).

Table 1		
Engine type and fuel	Maximum engine power	Carbon monoxide emission standard (g/hp-hr)
Non-emergency SI natural gas and non-emergency SI lean burn LPG	100≤hp<157	4.0
Landfill/digester gas	100≤hp <157	5.0

III.D.2.c.(ii)(C) Four stroke rich burn stationary reciprocating internal combustion engines with a site rating of more than 500 brake horsepower and located at a major HAP source are exempt from the carbon monoxide emissions limitations in Section III.D.2.c.(ii) and Table 1 but must comply with the formaldehyde emissions limitation in 40 CFR Part 63, Subpart ZZZZ, Table 1a (August 20, 2010).

III.D.1.c.(ii)(D) Compliance with the applicable carbon monoxide emissions limit must be achieved upon and after startup of the engine.

III.D.2.c.(iii) As an alternative to compliance with the carbon monoxide emission limit of 0.40 g/hp-hr in Section III.D.2.c.(ii), the owner or operator may demonstrate compliance with the following.

III.D.2.c.(iii)(A) For lean burn 2-stroke engines, a 58% reduction in carbon monoxide.

III.D.2.c.(iii)(B) For lean burn 4-stroke engines, a 93% reduction in carbon monoxide.

- III.D.2.c.(iii)(C) For rich burn engines, a 90% reduction in carbon monoxide.
- III.D.2.c.(iii)(D) The percent reduction is determined by testing according to the MACT ZZZZ testing requirements for carbon monoxide reduction in 40 CFR Part 63, Subpart ZZZZ, Section 63.6620 (August 30, 2024).
- III.D.2.c.(iv) The carbon monoxide emission limit of 0.40 g/hp-hr does not apply to periods of startup or shutdown. During periods of startup or shutdown, owners or operators must maintain and operate the engine in a manner consistent with good air pollution control practices for minimizing emissions.
- III.D.2.d. Beginning calendar year 2027 or the calendar year after initial startup, whichever is later, the owner or operator of a stationary reciprocating internal combustion engine at facility described in Sections III.D.1.a, III.D.1.b.(iii), or III.D.1.c.(iv)(B) where the III.D.1.c.(iv)(B) source also meets the criteria set forth in either Section III.D.1.c.(i) or III.D.1.c.(iii) must conduct the following inspections and adjustments annually.
 - III.D.2.d.(i) Change oil and filters as necessary.
 - III.D.2.d.(ii) Inspect air cleaners, fuel filters, hoses, and belts and clean or replace as necessary.
 - III.D.2.d.(iii) Inspect spark plugs and replace as necessary.
 - III.D.2.d.(iv) The owner or operator of a stationary reciprocating internal combustion engine that is subject to and required to conduct a periodic tune-up or combustion adjustment by the applicable requirements of a New Source Performance Standard in 40 CFR Part 60 (July 1, 2025) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (July 1, 2025) may conduct tune-ups or adjustments according to the schedule and procedures of the applicable requirements of 40 CFR Part 60 (July 1, 2025) or 40 CFR Part 63 (July 1, 2025).
- III.D.2.e. Performance testing and monitoring.

III.D.2.e.(i) The owner or operator of an engine subject to the emissions limitations requirements in Section III.D.2.c.(i), Table 1, or Section III.D.2.c.(ii)(C) must comply with the applicable performance testing requirements in 40 CFR Part 63, Subpart ZZZZ, Section 63.6620 (August 30, 2024) or 40 CFR Part 60, Subpart JJJJ, Section 60.4244 (August 30, 2024).

III.D.2.e.(ii) The owner or operator of an engine subject to the 0.4 g/hp-hr emissions limitation in Section III.D.2.c.(ii) must

III.D.2.e.(ii)(A) Conduct an initial performance test for carbon monoxide within 180 days after startup in accordance with the applicable reference test methods of 40 C.F.R. Part 60, Appendix A (July 1, 2019), and a test protocol submitted to the Division for review at least thirty (30) days prior to testing and in accordance with AQCC Common Provisions Regulation Section II.C.

III.D.2.e.(ii)(B) Conduct semi-annual portable analyzer monitoring for carbon monoxide, beginning within 180 days after completion of the initial performance test required in Section III.D.2.e.(ii)(A).

III.D.2.e.(ii)(B)(1) If the engine does not operate continuously, conduct portable analyzer monitoring for carbon dioxide semi-annually or within each 4,380-hour operating period.

III.D.2.e.(ii)(B)(2) If the engine does not operate continuously and does not exceed 4,380 hours of operation within two (2) calendar years, a minimum of one portable analyzer test must be completed every two years.

III.D.2.e.(ii)(B)(3) At least sixty (60) calendar days must separate the semi-annual tests.

III.D.2.e.(ii)(B)(4) All portable analyzer testing required by this section must be

conducted using the Division's Portable Analyzer Monitoring Protocol (version: March 2006).

III.D.2.e.(ii)(B)(5) Tuning of an engine prior to semi-annual monitoring events required by this Section III.D.2.e. is not a violation of this rule. However, readjustment of an engine set point following the monitoring event that would negatively impact the performance (i.e., result in increased emissions above applicable permit limits) of the engine is a violation of this rule.

III.D.2.e.(ii)(B)(6) A performance test conducted pursuant to Section III.D.2.e., 40 CFR Part 60, Subpart JJJJ (August 30, 2024), or a permit requirement may take the place of the next required semi-annual portable analyzer test required by this section.

III.D.2.e.(ii)(B)(7) An engine subject to at least semi-annual portable analyzer testing requirements in an existing permit issued by the Division can comply with this Section III.D.2.e. by complying with the testing requirements in the permit.

III.D.2.e.(iii) The owner or operator of an engine complying with the alternative emissions limitation in Section III.D.2.c.(iii) must

III.D.2.e.(iii)(A) Conduct an initial performance test for carbon monoxide within 180 days after startup.

III.D.2.e.(iii)(B) Conduct subsequent performance tests annually.

III.D.2.e.(iii)(C) Performance tests must be conducted in accordance with 40 CFR Part 63, Subpart ZZZZ, Section 63.6620 (August 30, 2024).

III.D.2.e.(iv) Tuning of an engine prior to a performance test required by Sections III.D.2.e.(ii) or III.D.2.e.(iii) is not a violation of this rule. However, readjustment of an engine set point following the performance test that would negatively impact the performance of the engine (i.e. result in increased emissions above applicable permit limits) is a violation of this rule.

III.D.2.f. Recordkeeping. The owner or operator of each spark ignition stationary reciprocating internal combustion engine subject to a requirement under this Section III.D.2. must maintain the following records, as applicable, for a period of five (5) years and make records available to the Division upon request.

III.D.2.f.(i) The applicable records in 40 CFR Sections 63.6655 (August 30, 2024) and 63.6660 (March 3, 2010) or in 40 CFR Section 60.4245 (August 30, 2024).

III.D.2.f.(ii) Records demonstrating compliance with Section III.D.2.d., including

III.D.2.f.(ii)(A) The date of the inspection.

III.D.2.f.(ii)(B) A description of what adjustment procedures were performed.

III.D.2.f.(ii)(C) A description of any corrective action taken.

III.D.2.f.(ii)(D) Whether the inspection was conducted pursuant to a New Source Performance Standard or National Emission Standard for Hazardous Air Pollutants, and which standard.

III.D.2.g. Reporting.

III.D.2.g.(i) The owner or operator of each source as described in Section III.D.1. must submit the applicable reports in 40 CFR Sections 63.6645 and 63.6650 (August 30, 2024) or in 40 CFR Section 60.4245 (August 30, 2024).

III.D.2.g.(ii) The owner or operator of a stationary reciprocating internal combustion engines subject to Sections III.D.2.c.(ii) or III.D.2.c.(iii), must submit

performance test results to the Division within thirty (30) days of completion of each test.

III.D.3. Requirements for Stationary Combustion Turbines.

III.D.3.a. Exemptions.

III.D.3.a.(i) Equipment or activities that are not required to be reported on TAC reports under Regulation Number 3, Part A, Section IX.A.2. or Regulation Number 7, Part B, Section V. are not subject to the requirements in Sections III.D.3.c. through III.D.3.h.

III.D.3.a.(ii) Stationary combustion turbines that commenced construction or reconstruction on or before September 1, 2026, if the owner or operator will remove or electrify the turbine as an emissions reduction measure to comply with Regulation Number 7, Part B, Section VII.

III.D.3.b. Definitions.

III.D.3.b.(i) "Area HAP source" means a stationary source of HAP that is not a major HAP source.

III.D.3.b.(ii) "Commenced construction" means for a stationary combustion turbine, the owner or operator has undertaken or has entered into a contractual obligation to undertake and complete a continuous program of construction. Construction meaning the on-site fabrication, erection, or installation of the stationary combustion turbine. Construction does not include the removal of a stationary combustion turbine from an existing location and reinstallation of such turbine at a new location.

III.D.3.b.(iii) "Deviation," for purposes of this Section III.D.3., means any instance in which a stationary combustion turbine fails to meet an emissions or operating limitation.

III.D.3.b.(iv) "Fixed capital cost" means the capital needed to provide all the depreciable components of an existing stationary combustion turbine.

III.D.3.b.(v) "Major HAP source" has the same meaning as defined in Regulation Number 3, Part A, Section I.B.

III.D.3.b.(vi) "Reconstruction" means the replacement of components of an affected or previously non-affected stationary combustion turbine to such extent that (1) the fixed capital cost of the new components exceeds fifty (50) percent of the fixed capital cost that would be required to construct a comparable new source and (2) it is technologically and economically feasible for the reconstructed source to meet the requirements in this Section III.D.3.

III.D.3.b.(vii) "Startup" begins at the first firing of fuel in the stationary combustion turbine. For simple cycle turbines, startup ends when the stationary combustion turbine has reached stable operation or after 1 hour, whichever is less. For combined cycle turbines, startup ends when the stationary combustion turbine has reached stable operation or after 3 hours, whichever is less. Turbines in combined cycle configurations that are operating as simple cycle turbines must meet the startup requirements for simple cycle turbines while operating as simple cycle turbines.

III.D.3.b.(viii) "Stationary combustion turbine" means all equipment, including but not limited to the turbine, the fuel, air, lubrication and exhaust gas systems, control systems (except emissions control equipment), and any ancillary components and sub-components comprising any simple cycle stationary combustion turbine, any regenerative/recuperative cycle stationary combustion turbine, the combustion turbine portion of any stationary cogeneration cycle combustion system, or the combustion turbine portion of any stationary combined cycle steam/electric generating system. Stationary means that the combustion turbine is not self-propelled or intended to be propelled when performing its function. Stationary combustion turbines do not include turbines located at a research or laboratory facility, if research is conducted on the turbine itself and the turbine is not being used to power other applications at the research or laboratory facility.

III.D.3.c. Emissions limitations.

III.D.3.c.(i) Beginning January 1, 2027, unless implementation date is otherwise specified in Section

III.D.1.g., the owner or operator of a stationary combustion turbine that commenced construction or reconstruction after January 14, 2003, at a stationary source described in Section III.D.1.a. that is also a major HAP source must comply with the formaldehyde emissions limitations in 40 CFR Part 63, Subpart YYYYY, Table 1 (March 9, 2022) and the operating limitations in 40 CFR Part 63, Subpart YYYYY, Table 2 (March 9, 2022) that correspond to the turbine.

III.D.3.c.(ii) The owner or operator of a stationary combustion turbine described in Sections III.D.3.c.(ii)(A) and III.D.3.c.(ii)(B) that is not subject to the formaldehyde emissions limitations in 40 CFR Part 63, Subpart YYYYY (March 9, 2022) must comply with either Section III.D.3.c.(ii)(C) or III.D.3.c.(ii)(D).

III.D.3.c.(ii)(A) A stationary combustion turbine that commences construction or reconstruction after September 1, 2026, at a stationary source described in Sections III.D.1.a. or a stationary source described in Section III.D.1.c.(iv)(B) that meets the criteria set forth in either Section III.D.1.c.(i) or III.D.1.c.(ii).

III.D.3.c.(ii)(B) A stationary combustion turbine that commences construction or reconstruction at a stationary source described in Section III.D.1.b.(iii).

III.D.3.c.(ii)(C) The owner or operator must comply with the applicable carbon monoxide emission limit provided in Table 2.

III.D.3.c.(ii)(C)(1) Compliance with the applicable carbon monoxide emissions limit must be achieved upon and after startup of the stationary combustion turbine.

Table 2			
Maximum heat input (MMBtu/hr)		CO emissions concentration (ppmvd @ 15% O ₂)	
		Gaseous fuels	Liquid fuels
≥ 10 and < 150	Simple and regenerative	6.0	10.0
≥ 150	Simple and regenerative	4.5	5.0
≥ 10 and < 150	Combined	5.0	10.0
≥ 150	Combined	2.0	2.0

III.D.3.c.(ii)(D) The owner or operator must comply with the formaldehyde emissions limitations in 40 CFR Part 63, Subpart YYYY, Table 1 (March 9, 2022).

III.D.3.c.(ii)(D)(1) Compliance with the applicable formaldehyde emissions limit must be achieved upon and after startup of the turbine.

III.D.3.c.(iii) The emissions limits in Table 2 do not apply during periods of startup or shutdown. During periods of startup or shutdown, owners or operators must maintain and operate the stationary combustion turbine in a manner consistent with good air pollution control practices for minimizing emissions.

III.D.3.d. All stationary combustion turbines and associated air pollution control and monitoring equipment located at a stationary source described in Sections III.D.1.a., III.D.1.b.(iii), or III.D.1.c.(iv)(B) where the III.D.1.c.(iv)(B) source also meets the criteria set forth in either Section III.D.1.c.(i) or III.D.1.c.(iii) must be operated and maintained pursuant to the manufacturing specifications or equivalent to

the extent practicable, and consistent with technological limitations and good engineering and maintenance practices. The owner or operator must keep manufacturer specifications or equivalent on file.

III.D.3.e. Compliance monitoring and performance testing.

III.D.3.e.(i) The owner or operator of a stationary combustion turbine subject to Section III.D.3.c.(i) must comply with the performance testing requirements in 40 CFR Part 63, Subpart YYYYY, Table 3 (March 9, 2022).

III.D.3.e.(ii) The owner or operator of a stationary combustion turbine greater than or equal to 150 MMBtu/hr complying with Section III.D.3.c.(ii)(C) must demonstrate compliance with the applicable carbon monoxide emissions limit in Table 2 using 3-hour rolling averaging times based on 1-hour blocks as monitored by a continuous emissions monitoring system (CEMS).

III.D.3.e.(iii) The owner or operator of a stationary combustion turbine less than 150 MMBtu/hr complying with Section III.D.3.c.(ii)(C) must demonstrate compliance with the applicable carbon monoxide emissions limit in Table 2 using

III.D.3.e.(iii)(A) An initial performance test for carbon monoxide within 180 days after startup conducted in accordance with the applicable reference test methods of 40 C.F.R. Part 60, Appendix A (July 1, 2019), and a test protocol submitted to the Division for review at least thirty (30) days prior to testing and in accordance with AQCC Common Provisions Regulation Section II.C., and

III.D.3.e.(iii)(B) Quarterly portable analyzer monitoring for carbon monoxide.

III.D.3.e.(iii)(B)(1) For stationary combustion turbines that do not operate continuously, conduct portable analyzer monitoring quarterly or within each 2,190 hour operating period.

III.D.3.e.(iii)(B)(2) For stationary combustion turbines that do not operate continuously and that do not exceed 2,190 hours of operation within two years, a minimum of one portable analyzer test must be completed every two years.

III.D.3.e.(iii)(B)(3) At least 30 days must separate the quarterly tests.

III.D.3.e.(iii)(C) Portable analyzer testing must be conducted using the Division's Portable Analyzer Monitoring Protocol (version: March 2006).

III.D.3.e.(iii)(D) A performance test conducted pursuant to 40 CFR Part 60, Subparts KKKK or KKKKa (January 15, 2026), or a permit requirement may take the place of the next required quarterly portable analyzer test required by this section.

III.D.3.e.(iii)(E) Tuning of a turbine prior to quarterly monitoring events or performance tests required by this Section III.D.3.e. is not a violation of this rule. However, readjustment of a turbine set point following the monitoring event that would negatively impact the performance (i.e., result in increased emissions above applicable permit limits) of the turbine is a violation of this rule.

III.D.3.e.(iii)(F) As an alternative to conducting performance tests and quarterly portable analyzer monitoring, the owner or operator may demonstrate compliance with the carbon monoxide emissions limit using 3-hour rolling averaging times based on 1-hour blocks as monitored by a continuous emissions monitoring system (CEMS).

III.D.3.e.(iv) The owner or operator of a stationary combustion turbine complying with Section III.D.3.c.(ii)(D) must

- III.D.3.e.(iv)(A) Conduct an initial performance test for formaldehyde within 180 days after startup.
- III.D.3.e.(iv)(B) Conduct subsequent performance tests for formaldehyde annually.
- III.D.3.e.(iv)(C) Performance tests must be conducted in accordance with the procedures in 40 CFR, Part 63, Subpart YYYY, Table 3 (March 9, 2022).
- III.D.3.e.(iv)(D) As an alternative to conducting performance tests, the owner or operator may demonstrate compliance with the formaldehyde emissions limitation using 3-hour rolling averaging times based on 1-hour blocks as monitored by a continuous emissions monitoring system (CEMS).
- III.D.3.e.(v) The owner or operator of a stationary combustion turbine utilizing an oxidation catalyst to comply with the applicable carbon monoxide or formaldehyde emissions limitation in Sections III.D.3.c.(ii)(C) or III.D.3.c.(ii)(D), and that is not demonstrating compliance using a CEMS pursuant to Sections III.D.3.e.(ii), III.D.3.e.(iii)(F) or III.D.3.e.(iv)(D), must continuously monitor the inlet temperature to the catalyst and maintain the 4-hour rolling average of the inlet temperature within the range suggested by the catalyst manufacturer. Catalyst inlet temperature data recorded during startup is not required to be included in the calculations of the 4-hour rolling average catalyst inlet temperature.
- III.D.3.e.(vi) The owner or operator of a stationary combustion turbine not utilizing an oxidation catalyst to comply with the formaldehyde emissions limitation in Section III.D.3.c.(ii)(D) or the carbon monoxide emissions limitation in Section III.D.3.c.(ii)(C) and compliance is not demonstrated as monitored by a continuous emissions monitoring system (CEMS) must within 30 days after completion of the initial compliance test develop and implement an operating parameter monitoring plan that includes operating parameters and ranges to demonstrate compliance

with the applicable emissions limitation. The operating plan must

III.D.3.e.(vi)(A) Identify the operating parameters that influence formaldehyde or carbon monoxide emissions for the specific stationary combustion turbine, which may include parameters such as combustion chamber temperature, turbine load, air-to-fuel ratio, fuel characteristics, or any other parameters demonstrated to be reliable indicators of emission performance;

III.D.3.e.(vi)(B) Establish allowable operating ranges for each selected parameter based on the conditions measured during the most recent performance test and any manufacturer recommendations as applicable, including documentation of how the ranges were derived and justification that the ranges ensure ongoing compliance with the applicable emission limit;

III.D.3.e.(vi)(C) Describe the monitoring equipment, measurement methods, and monitoring frequency for each parameter, including calibration, quality assurance, and quality control procedures sufficient to ensure data accuracy, as applicable;

III.D.3.e.(vi)(D) Specify the corrective actions the owner or operator will take when any monitored parameter falls outside the established allowable range, including procedures for returning the stationary combustion turbine to compliant operation and documenting the duration and cause of the excursion; and

III.D.3.e.(vi)(E) Include recordkeeping and reporting procedures that document parameter readings, excursions, corrective actions, and any updates to the operating parameter monitoring plan.

III.D.3.e.(vi)(F) The owner or operator must maintain the operating monitoring plan and

revise the plan whenever a performance test, equipment modification, or operational change affects the parameters or ranges relied upon to demonstrate compliance. Plans must be made available to the Division upon request. Revisions must be made to the plans at the request of the Division.

III.D.3.f. Beginning in calendar year 2027 or the calendar year after initial startup, whichever is later, the owner or operator of a stationary combustion turbine at a facility described in Sections III.D.1.a., III.D.1.b.(iii), or III.D.1.c.(iv)(B) where the III.D.1.c.(iv)(B) source also meets the criteria set forth in either Section III.D.1.c.(i) or III.D.1.c.(iii) must conduct the following inspections and adjustments annually.

III.D.3.f.(i) Inspect turbine inlet systems and align, repair, or replace components as necessary.

III.D.3.f.(ii) Inspect the combustion chamber components, combustion liners, transition pieces, and fuel nozzle assemblies and clean, repair, or replace components as necessary.

III.D.3.f.(iii) Inspections must be conducted when burning the fuel that provides the majority of the heat input since the last combustion process adjustment and when operating at a firing rate typical of normal operation, confirm proper setting and calibration of the combustion controls.

III.D.3.f.(iv) The owner or operator of a stationary combustion turbine that is subject to and required to conduct a periodic tune-up or combustion adjustment by the applicable requirements of a New Source Performance Standard in 40 CFR Part 60 (July 1, 2025) or National Emission Standard for Hazardous Air Pollutants in 40 CFR Part 63 (July 1, 2025) may conduct tune-ups or adjustments according to the schedule and procedures of the applicable requirements of 40 CFR Part 60 (July 1, 2025) or 40 CFR Part 63 (July 1, 2025).

III.D.3.f.(v) If the stationary combustion turbine was constructed for the purpose of supplying more than one-third of its potential electric output capacity to any

utility power distribution system for sale, the inspection may be delayed

III.D.3.f.(v)(A) Until a unit outage, not to exceed twenty-four (24) months from the prior inspection required by Section III.D.3.f.; or

III.D.3.f.(v)(B) If the stationary combustion turbine is offline when the deadline to perform the inspection required by Section III.D.3.f. passes. The owner or operator of the stationary combustion turbine must perform the inspection within 720 unit operating hours after startup.

III.D.3.g. Recordkeeping. The owner or operator of each stationary combustion turbine subject to a requirement in this Section III.D.3. must maintain the following records, as applicable, for a period of five (5) years and make records available to the Division upon request.

III.D.3.g.(i) For stationary combustion turbines subject to Section III.D.3.c.(i), the applicable records in 40 CFR Section 63.6155 (March 9, 2022).

III.D.3.g.(ii) For stationary combustion turbines subject to Section III.D.3.c.(ii), records of, as applicable

III.D.3.g.(ii)(A) Performance tests;

III.D.3.g.(ii)(B) The date, time, and duration of each startup;

III.D.3.g.(ii)(C) Daily fuel usage for lean premix gas-fired stationary combustion turbines or diffusion flame gas-fired stationary combustion turbines that use any quantity of distillate oil;

III.D.3.g.(ii)(D) Maintenance on the air pollution control equipment;

III.D.3.g.(ii)(E) Records of manufacturer guarantee and owner or operator operation and maintenance in accordance with manufacturer instructions for the continuous emissions monitoring system (CEMS), if required pursuant to Section III.D.3.e.(ii) or used pursuant to III.D.3.e.(iii)(F);

- III.D.3.g.(ii)(F) Records to show compliance with the catalyst inlet temperature as specified in Section III.D.3.e.(v);
- III.D.3.g.(ii)(G) Records of the operating plan as specified in Section III.D.3.e.(vi); and
- III.D.3.g.(ii)(H) Records of manufacturer specifications or equivalent related to compliance with the operation and maintenance requirement in Section III.D.3.d.
- III.D.3.g.(iii) Records demonstrating compliance with Section III.D.3.f., including
 - III.D.3.g.(iii)(A) The date of the inspection.
 - III.D.3.g.(iii)(B) A description of what adjustment procedures were performed.
 - III.D.3.g.(iii)(C) A description of any corrective action taken.
 - III.D.3.g.(iii)(D) Whether the inspection was conducted pursuant to a New Source Performance Standard or National Emission Standard for Hazardous Air Pollutants, and which standard.
- III.D.3.g.(iv) Records of CEMS monitoring, as applicable.
- III.D.3.g.(v) Records of the number of deviations, including
 - III.D.3.g.(v)(A) The date, time, cause, and duration of the deviation;
 - III.D.3.g.(v)(B) A list of the affected stationary combustion turbine(s), an estimate of the quantity of each regulated pollutant emitted over any emission limitation and a description of the method used to estimate the emissions; and
 - III.D.3.g.(v)(C) Description of actions taken to minimize emissions and any corrective actions taken to return the affected stationary combustion turbine to normal operation.

III.D.3.h. Reporting. The owner or operator of each stationary combustion turbine subject to a requirement in this Section III.D.3. must submit the following reports to the Division, as applicable.

III.D.3.h.(i) For stationary combustion turbines at sources specified in Section III.D.3.c.(i), the applicable reports in 40 CFR Sections 63.6145 and 63.6150 (March 9, 2022).

III.D.3.h.(ii) For stationary combustion turbines subject to Section III.D.3.c.(ii), owners or operators must submit the following annually on a Division-approved form,

III.D.3.h.(ii)(A) Company name and address;

III.D.3.h.(ii)(B) Date of the report and beginning and ending dates of the reporting period;

III.D.3.h.(ii)(C) Identification of each stationary combustion turbine included in the report, including AIRS number;

III.D.3.h.(ii)(D) Lean premix gas-fired stationary combustion turbines or diffusion flame gas-fired stationary combustion turbines that use any quantity of distillate oil must include the number of hours distillate oil was fired during the reporting period;

III.D.3.h.(ii)(E) For stationary combustion turbines determining compliance using CEMS, semi-annual excess emissions reports no later than the 30th day following the end of each semi-annual period. Excess emissions means emissions that exceed the applicable limitations contained in Section III.D.3.b.(ii). Excess emission reports must include the information specified in 40 CFR Part 60, Section 60.7(c) (February 12, 1999); and

III.D.3.h.(ii)(F) For each deviation,

III.D.3.h.(ii)(F)(1) The start date, start time, duration, and cause of each deviation, and the corrective action taken;

III.D.3.h.(ii)(F)(2) A list of the affected stationary combustion turbine(s), an estimate of the quantity of each regulated pollutant emitted over any emission limit, and a description of the method used to estimate the emissions;

III.D.3.h.(ii)(F)(3) Information on the number, duration, and cause for monitor downtime incidents (including unknown cause, if applicable, other than downtime associated with zero and span and other daily calibration checks), as applicable, and the corrective action taken; and

III.D.3.h.(ii)(F)(4) The total operating time of the stationary combustion turbine during the reporting period.

III.D.3.h.(ii)(G) Each report must be accompanied by a certification by a responsible official that, based on the information and belief formed after reasonable inquiry, the statements and information in the document are true, accurate, and complete.

III.E. Hydrogen sulfide

III.E.1. Applicability

III.E.1.a. An owner or operator of a stationary source that (1) commenced construction before June 14, 2026; (2) is an asphalt processing plant, has asphalt roofing plant operations, or operates an anaerobic digester; and (3) has actual emissions of hydrogen sulfide at or above 10,000 pounds per year (lb/yr).

III.E.1.a.(i) The stationary source's actual annual hydrogen sulfide emissions are determined as follows:

III.E.1.a.(i)(A) The highest stationary source-wide actual emissions of hydrogen sulfide recorded on the stationary source's 2023, 2024, or 2025 TAC report (as available). If the stationary source did not submit a 2023, 2024, or 2025 TAC report, the emissions are based on the

most recent Air Pollutant Emission Notices (APENs), excluding any cancelled APEN for points that were not operating after June 30, 2026, data (if available), unless an emissions update report is received by June 30, 2026.

III.E.1.a.(i)(B) A stationary source may choose to determine actual emissions of hydrogen sulfide based on the three year average of the actual emissions of hydrogen sulfide recorded on the stationary source's TAC reports for calendar years 2023, 2024, and 2025.

III.E.1.a.(i)(C) A stationary source that commenced construction prior to June 14, 2026, but did not commence operation as of June 14, 2026, must determine actual emissions of hydrogen sulfide based on the first 12 months after commencing operation.

III.E.1.b. An owner or operator of a stationary source that commences construction on or after June 14, 2026, that will operate an asphalt processing plant, asphalt roofing plant operation, or an anaerobic digester, with forecasted actual hydrogen sulfide emissions must submit the forecasted hydrogen sulfide emissions as specified in Sections III.E.1.b.(i) through III.E.1.b.(iii).

III.E.1.b.(i) Forecasted actual hydrogen sulfide emissions must be submitted in a Division-approved format with the permit application.

III.E.1.b.(ii) If no permit application is required, the forecasted actual hydrogen sulfide emissions must be submitted to the Division in a Division-approved format before the source commences operation of the asphalt processing plant, asphalt roofing plant operation, or anaerobic digester.

III.E.1.b.(iii) If the forecasted actual hydrogen sulfide emissions meet or exceed ten thousand (10,000) pounds per year (lb/yr), the owner or operator of such stationary source must comply with relevant emission control requirements in this Section III.E. upon commencement of operation of the asphalt processing plant, asphalt roofing plant operation, or anaerobic digester.

- III.E.1.c. An owner or operator of a stationary source that commences construction before June 14, 2026, that undergoes a modification of an asphalt processing plant, asphalt roofing plant operation, or anaerobic digester after June 14, 2026, must submit the forecasted actual hydrogen sulfide emissions as specified in Sections III.E.1.c.(i) through III.E.1.c.(iii).
 - III.E.1.c.(i) Forecasted actual hydrogen sulfide emissions must be submitted in a Division-approved format with the permit application.
 - III.E.1.c.(ii) If no permit application is required, the forecasted actual hydrogen sulfide emissions must be submitted to the Division in a Division-approved format before the source commences construction or reconstruction on the asphalt processing plant, asphalt roofing plant operation, or anaerobic digester.
 - III.E.1.c.(iii) If the forecasted actual hydrogen sulfide emissions meet or exceed ten thousand (10,000) pounds per year (lb/yr), the owner or operator of such stationary source must comply with relevant emission control requirements in this Section III.E. upon commencement of operation of the asphalt processing plant, asphalt roofing plant operation, or anaerobic digester.
- III.E.1.d. Emissions from equipment or activities that are not required to be reported on TAC reports under Regulation Number 3, Part A, Section IX.A.2. (or for ONGAEIR sources: activities, emissions, or equipment not required to be reported under Regulation Number 7, Part B, Section V.), will not be considered in determining whether the stationary source-wide emissions meet or exceed the hydrogen sulfide emission threshold in this Section III.E.1.
- III.E.1.e. Definitions, for purposes of determining applicability.
 - III.E.1.e.(i) “Actual emissions” (for the purposes of this Part B, Section III.), has the same meaning set forth in Regulation Number 3, Part A, Section IX.
 - III.E.1.e.(ii) “Commenced construction” for a stationary source, has the same meaning set forth in Regulation Number 3, Part A, Section I.B.14.

III.E.1.e.(iii) “Forecasted actual hydrogen sulfide emissions” means the highest actual emissions, in tons per year, at which a stationary source described in Sections III.E.1.b. or III.E.1.c. is projected to emit hydrogen sulfide in any one of the five calendar years (twelve-month periods) including and following the commencement of operation of the new or modified stationary source. In determining the forecasted actual hydrogen sulfide emissions, before commencing construction, the owner or operator of the stationary source must consider all relevant information, including but not limited to, historical operational data, the company's own representations, the company's expected business activity and the company's highest projections of business activity, the company's filings with the State or Federal regulatory authorities, and compliance plans.

III.E.1.e.(iv) “Modified” or “modification” has the same meaning set forth in Regulation Number 3, Part A, Section I.B.33.

III.E.1.e.(v) “Stationary source” has the meaning set forth in Regulation Number 3, Part A, Section I.B.52.

III.E.1.f. Recordkeeping, for the purposes of determining applicability.

The owner or operator of each stationary source as described in Sections III.E.1. must maintain the following records, as applicable, for a period of five (5) years and make records available to the Division upon request.

III.E.1.f.(i) No later than June 30, 2026, an owner or operator of a stationary source that is determining applicability in accordance with Section III.E.1.a. must maintain the following records.

III.E.1.f.(i)(A) Records of the stationary source's actual emissions of hydrogen sulfide and the supporting methodology used to determine emissions in accordance with Section III.E.1.a.(i), and the date of the initial emissions threshold determination.

III.E.1.f.(ii) No later than 60 days after the first 12 calendar months of operation, an owner or operator of a

stationary source that commenced construction prior to June 14, 2026, but did not commence operation as of June 14, 2026, and that determines stationary source applicability in accordance with Section III.E.1.a.(i)(C) must maintain records of stationary source-wide actual emissions of hydrogen sulfide.

III.E.1.f.(iii) An owner or operator that is determining stationary source applicability in accordance with Section III.E.1.b. must maintain records of stationary source-wide forecasted actual hydrogen sulfide emissions and the date of the initial applicability emission threshold determination.

III.E.1.f.(iii)(A) An owner or operator that is determining applicability in accordance with Section III.E.1.b., its forecasted actual hydrogen sulfide emissions are below the 10,000 lb/yr applicability threshold must maintain records of the stationary source-wide annual actual emissions of hydrogen sulfide for each of the five (5) calendar years after commencement of operation of the new stationary source.

III.E.1.f.(iv) An owner or operator that is determining stationary source applicability in accordance with Section III.E.1.c., must maintain records of stationary source-wide forecasted actual hydrogen sulfide emissions and the date of the initial applicability emission threshold determination.

III.E.1.f.(iv)(A) An owner or operator that is determining applicability in accordance with Section III.E.1.c., its forecasted actual hydrogen sulfide emissions are below the 10,000 lb/yr applicability threshold must maintain records of the stationary source-wide annual actual emissions of hydrogen sulfide for each of the five (5) calendar years after commencement of operation of the modified stationary source.

III.E.1.f.(v) Records of reports related to applicability submitted pursuant to Section III.E.1.h.

III.E.1.g. Reports related to applicability.

III.E.1.g.(i) Emission update report: No later than June 30, 2026, if an owner or operator assessing stationary source-wide emissions in accordance with Section III.E.1.a. elects to use the emission update mechanism in Section III.E.1.a.(i)(A) and stationary source-wide actual emissions of hydrogen sulfide are below 10,000 pounds per year (lb/yr), the owner or operator must submit an update to previously reported emissions in a Division-approved format.

III.E.1.g.(i)(A) The emissions update must verify that the stationary source-wide actual emissions are below the hydrogen sulfide emissions threshold of 10,000 pounds per year (lb/yr) based on calendar year 2025 actual emissions, or based on the three year average of the actual emissions of hydrogen sulfide recorded on the stationary source's TAC reports for calendar years 2023, 2024, and 2025, in accordance with procedures and exemptions set forth in Regulation Number 3, Part A, Section IX., or Regulation Number 7, Part B, Section V.

III.E.1.g.(ii) Notice of Applicability for an existing stationary source that had not yet commenced operation as of June 14, 2026: No later than 60 days after the first 12 months of operation, for an owner or operator of a stationary source that commenced construction prior to June 14, 2026, but had not yet commenced operation as of that date, and the stationary source-wide actual emissions of hydrogen sulfide meet or exceed 10,000 pounds per year (lb/yr), the owner or operator must:

III.E.1.g.(ii)(A) Submit a notification of applicability to the Division which verifies the calendar year of actual emissions of hydrogen sulfide from the first 12 months after commencing operation met or exceeded 10,000 pounds per year (lb/yr); and

III.E.1.g.(ii)(B) Comply with any applicable requirements in this Section III.E. within one hundred and eighty (180) days of the date the notification was due.

III.E.1.g.(iii) Notice of Applicability for a new stationary source: If a stationary source's forecasted actual hydrogen sulfide emissions are below 10,000 pounds per year (lb/yr), and the source subsequently has actual emissions of hydrogen sulfide at or above 10,000 lb/yr in any of the first five (5) calendar years after operation, no later than March 1 of the subsequent calendar year where actual emissions of hydrogen sulfide meet or exceed 10,000 pounds per year (lb/yr), the owner or operator must submit a notification of applicability in a Division-approved format which verifies the calendar year of actual emissions of hydrogen sulfide that met or exceeded 10,000 pounds per year (lb/yr).

III.E.1.g.(iii)(A) The owner or operator of such stationary source must comply with any applicable requirements in this Section III.E. within one hundred and eighty (180) days of the date the notification was due.

III.E.1.g.(iv) Notice of Applicability after a stationary source modifies: If a stationary source's forecasted actual hydrogen sulfide emissions are below 10,000 pounds per year (lb/yr), and the source subsequently has actual emissions of hydrogen sulfide at or above 10,000 lb/yr in any of the first five (5) calendar years after operation, no later than March 1 of the subsequent calendar year where actual emissions of hydrogen sulfide meet or exceed 10,000 pounds per year (lb/yr), the owner or operator must:

III.E.1.g.(iv)(A) Submit a notification of applicability to the Division which verifies the calendar year of actual emissions of hydrogen sulfide that met or exceeded 10,000 pounds per year (lb/yr); and

III.E.1.g.(iv)(B) Comply with any applicable requirements in this Section III.E. within one hundred and eighty (180) days of the date the notification was due.

III.E.2. Exemptions for categorical hydrogen sulfide requirements

III.E.2.a. The owner or operator of a stationary source that is an asphalt processing plant or has asphalt roofing plant

operations and that has submitted a permit application seeking a source-wide hydrogen sulfide emission limit below 10,000 pounds per year that has been received and deemed administratively complete by the Division as of June 14, 2026, is exempt from Section III.E., provided that:

III.E.2.a.(i) The owner or operator cooperates with the Division to obtain permit(s) limiting hydrogen sulfide emissions to below 10,000 pounds per year;

III.E.2.a.(ii) The owner or operator begins construction on any control devices necessary to meet the permitted limit(s) within 6 months of permit issuance; and

III.E.2.a.(iii) The stationary source's actual hydrogen sulfide emissions, after commencing operation of the control devices authorized by the final permit, remain below 10,000 pounds per year.

III.E.2.b. Equipment or activities that are not required to be reported on TAC reports under Regulation Number 3, Part A, Section IX.A.2. or Regulation Number 7, Part B, Section V. are not subject to the requirements in Section III.E.

III.E.3. Definitions for categorical hydrogen sulfide requirements

III.E.3.a. "Anaerobic digester" means a biological treatment system that decomposes organic materials such as manure anaerobically in the absence of oxygen to produce biogas. The generated biogas is made up of methane (CH₄), carbon dioxide (CO₂), and digestate, a nutrient-rich byproduct. During this process, sulfur-containing compounds in the feedstock are broken down by sulfate-reducing bacteria, generating H₂S as a byproduct.

III.E.3.b. "Asphalt processing plant" has the same meaning as 40 CFR Part 60, Subpart UU, Section 60.471 (October 17, 2000).

III.E.3.c. "Asphalt processing operation" has the same meaning as 40 CFR Part 63, Subpart AAAAAAA, Section 63.11556 (November 19, 2020).

III.E.3.d. "Asphalt roofing plant" has the same meaning as 40 CFR, Part 60, Subpart UU, Section 60.471 (October 17, 2000).

III.E.3.e. “Asphalt roofing manufacturing operation” has the same meaning as 40 CFR Part 63, Subpart AAAAAAA, Section 63.11556 (November 19, 2020).

III.E.4. Emission limitations

III.E.4.a. Asphalt processing plants and asphalt roofing plant operations. Beginning March 15, 2027, or upon commencement of operation if such date is after June 14, 2026, or upon such date as specified in Section III.E.1.g., owners or operators of a stationary source described in Sections III.E.1.a. or III.E.1.b. or III.E.1.c. with asphalt processing plant or asphalt roofing plant operations must utilize an abatement device (e.g., regenerative thermal oxidizer(s)) to control hydrogen sulfide emissions from asphalt manufacturing operations including such emissions from any tanks used for storage, surge, or mixing; coating or applicator equipment; sealant; and any loadout racks.

III.E.4.a.(i) The owner or operator must install and operate an emissions abatement device designed to achieve either a minimum hydrogen sulfide destruction efficiency of 97% or a minimum VOC destruction efficiency of 97%.

III.E.4.a.(ii) The owner or operator must conduct initial performance testing to demonstrate the destruction efficiency of either hydrogen sulfide or VOCs within 180 days of commencement of operation of the abatement device. If the owner or operator elects to demonstrate compliance via VOC destruction efficiency, such results will serve as a surrogate for hydrogen sulfide control performance. The pollutant selected for testing must be identified in a test protocol submitted to the Division for approval before conducting the test.

III.E.4.a.(iii) The owner or operator must maintain the thermal oxidizer's, or other abatement device's, 3-hour average combustion zone temperature at or above the operating value established during the most recent emission test as specified in 40 CFR Part 63, Subpart AAAAAAA, Table 4 (March 18, 2010).

Any installed abatement device must be operated in accordance with the manufacturer's specifications and a Division approved operation and maintenance plan.

III.E.5. Work practices

III.E.5.a. Asphalt processing plants and asphalt roofing plant operations. Beginning June 14, 2026, or upon commencement of operation if such date is after June 14, 2026, or upon such date as specified in Section III.E.1.g., owners or operators of a stationary source described in Sections III.E.1.a or III.E.1.b. or III.E.1.c., with asphalt processing operations or asphalt roofing manufacturing operations must comply with the following best management practices to minimize hydrogen sulfide emissions from asphalt processing operations or asphalt roofing manufacturing operations.

III.E.5.a.(i) Upon commencement of operation of the abatement device required under Section III.E.4.a., conduct inspection and maintenance as recommended by the manufacturer.

III.E.5.a.(ii) Conduct weekly visual inspections of asphalt processing or asphalt roofing manufacturing operations and perform maintenance on equipment as necessary.

III.E.5.a.(iii) Inspect and replace filters on asphalt processing or asphalt roofing manufacturing operations in accordance with the manufacturer's specifications and a Division-approved operation and maintenance plan (e.g., monthly, yearly).

III.E.5.a.(iv) Conduct and record inspection and maintenance on the asphalt storage and surge tanks, as follows:

III.E.5.a.(iv)(A) Daily visual inspections.

III.E.5.a.(iv)(B) Conduct routine inspections and cleaning; however, the interval between cleanings must not exceed five years.

III.E.5.a.(iv)(C) Perform structural thickness measurements in accordance with current manufacturer's specifications and schedule.

III.E.5.b. Anaerobic digesters. Beginning June 14, 2026, unless implementation date is otherwise specified in Section III.E.1.g., owners or operators of a stationary source described in Sections III.E.1.a or III.E.1.b. or III.E.1.c. with

anaerobic digesters that use manure as a feedstock for the digester must comply with the following best management practices to minimize emissions of hydrogen sulfide.

- III.E.5.b.(i) The owner or operator must continuously inject a non-corrosive hydrogen sulfide-control additive into the manure handling or storage system during transfer operations. The injection rate must be operationally optimized based on the manure flow rate to maintain hydrogen sulfide concentrations at or below 500 ppm prior to the abatement device. The owner or operator must submit a proposed operation and maintenance plan for Division approval that defines, at a minimum, the specific parameters for additive selection, injection points, and flow-rate optimization.
- III.E.5.b.(ii) During periods of system startup and, if there are malfunctions that delay the movement of the manure from the uncovered receiving pit or if uncovered storage pits will hold manure for longer than 1 day prior to moving into the covered storage tanks, non-corrosive hydrogen sulfide-control additive must also be added to these pit areas at a rate to be determined based on the volume of the manure in the pits.
- III.E.5.b.(iii) Regularly remove manure from the manure truck unloading pit or to the manure storage pit before they are routed to the manure storage tanks as often as possible but at a minimum of once per day to limit biological and chemical conditions favoring hydrogen sulfide formation and minimize fugitive hydrogen sulfide release.
- III.E.5.b.(iv) During periods of tank maintenance or cleaning, deliveries to the receiving pit must be halted only if the remaining operational storage capacity is insufficient to contain incoming volumes. Deliveries must be halted if all storage tanks are out of service for cleaning for a period exceeding 24 hours.
- III.E.5.b.(v) If there is a malfunction or shutdown of the digesters or storage tank and pump that prevent the manure from being moved out of the uncovered truck unloading pit or storage pits and into the covered

storage tanks for more than 1 day, deliveries must be halted until normal operations can resume.

- III.E.5.b.(vi) Maintain negative pressure and prevent gas buildup and corrosion in the manure storage tank(s) by interconnecting the vapor head space of the tank(s) to the nearest covered anaerobic digester.
- III.E.5.b.(vii) Maintain the inflated covers on the anaerobic digester tank(s) that collects and stores digester gas prior to transfer to the biogas transfer blower(s) or compressor station(s) in a manner consistent with recommendations by the manufacturer and with the requirements of the Division-approved operations and maintenance plan.
- III.E.5.b.(viii) Operate and maintain approved filtration equipment per the manufacturers' recommendation.
- III.E.5.b.(ix) Maintain a cover on the anaerobic lagoon system.
- III.E.5.b.(x) Operate and maintain the flare per the manufacturers' recommendation.
- III.E.5.b.(xi) Route all gases from the anaerobic digester tanks to the biogas upgrade system or to a flare. Following the upgrade process, the resulting gas must be injected into the natural gas pipeline or, if the gas is off-spec or the pipeline is unavailable, routed to a flare.
- III.E.5.b.(xii) Continuous hydrogen sulfide monitors must be operated and maintained in accordance with the manufacturer's recommendations and the Division-approved operation and maintenance plan at the fenceline to monitor fugitive emissions. Failure to operate a monitor according to these requirements must be reported to the Division within an approved timeframe.
- III.E.5.b.(xiii) The owner or operator must install and operate a Division approved continuous emission monitoring system or process analysis system, combined with a gas flow meter to measure concentrations of hydrogen sulfide (H₂S) from the anaerobic digester outlet/H₂S scrubber. Any installed continuous

emission monitoring system or process analysis system must be operated in accordance with the manufacturer's recommendations and a Division-approved operation and maintenance plan. Additionally, any installed process analysis system must undergo Quality Assurance/Quality Control (QA/QC) testing following the guidelines in the latest approved version of the "Guidelines for State-Only Required Continuous Monitoring Systems in the State of Colorado."

III.E.6. Recordkeeping. The following records must be kept for a period of five (5) years and made available to the Division upon request.

III.E.6.a. Records demonstrating compliance with the applicable emissions limitation requirements in Section III.E.4., as well as records of all maintenance and repairs conducted.

III.E.6.b. Records demonstrating compliance with the applicable best management practices in Section III.E.5.

III.E.6.c. Records of all versions of operations and maintenance plans and any reports or performance tests required under Section III.E.7.

III.E.7. Reporting

III.E.7.a. Beginning June 30, 2027, and annually thereafter, or within one year of commencing operation, or modifying, if after June 14, 2026, and annually thereafter, owners or operators of stationary sources described in Sections III.E.1.a., III.E.1.b., or III.E.1.c. with anaerobic digesters incorporating manure as a feedstock must submit an annual compliance report that covers the previous calendar year the following in a Division-approved format. The report must include the following:

III.E.7.a.(i) Company Information: Include the facility name, address, owner or operator name.

III.E.7.a.(ii) Any deviations from the required H₂S ppm value noted in Section III.E.5.b.(i), including the date of the occurrence, the duration of the occurrence, the cause of the occurrence, explanation of any corrective

actions, and a copy of the record(s) where the deviation was recorded.

III.E.7.a.(iii) The total number of startup or malfunction occurrences, and the duration of each such occurrence, that prevented the manure from being routed to the storage tanks for periods longer than noted in Sections III.E.5.b.(ii) and III.E.6.b.(iii). For each occurrence, provide a discussion of any corrective actions taken to resolve these operational concerns. This discussion shall include an indication of any precautions taken during those periods to minimize fugitive emissions through either delivery adjustments or additional pit additives.

III.E.7.a.(iv) The number and dates of any readings of fugitive monitors that exceed the alarm levels noted in the latest approved operations and maintenance plan, as well as an explanation of any corrective actions.

III.E.7.a.(v) An explanation of any procedures from the stationary source's operations and maintenance plan that were not followed during the reporting period and any corrective actions taken.

III.E.7.a.(vi) A certification that the procedures in the latest Division-approved operations and maintenance plan were followed, if applicable.

III.E.7.b. Beginning June 30, 2027, and annually thereafter, or within one year of commencing operation if after June 14, 2026, and annually thereafter, owners or operators of stationary sources described in Sections III.E.1.a or III.E.1.b. with asphalt processing plants, asphalt processing operations, asphalt roofing plants, or asphalt roofing manufacturing operations must submit an annual compliance report that covers the previous calendar year the following in a Division-approved format. The report must include the following:

III.E.7.b.(i) Company Information: Include the facility name, address, owner or operator name.

III.E.7.b.(ii) Any deviations from the destruction efficiency or operational parameters monitored under Section III.E.4.a.

III.F. Non-Categorical Existing Stationary Source Requirements

III.F.1. Non-Categorical Reporting Requirements

III.F.1.a. METAL CONTAINER CORP (AIRS ID: 123-0134), must submit the analysis in Section III.F.2. or, no later than September 1, 2026, submit a detailed emissions report on a Division approved form to the Division that demonstrates the stationary source's actual emissions, as of calendar year 2025, are below 4,000 lb/yr of formaldehyde or a certification that the stationary source's location does not meet either of the location criteria noted in Sections III.D.1.a.(i) or III.D.1.a.(ii).

III.F.1.b. No later than September 1, 2026, CF & I STEEL L.P. DBA EVRAZ ROCKY MTN STEEL (AIRS ID: 101-0048) must submit a detailed emissions report on a Division approved form using a Division-approved testing method to update its formaldehyde emission estimates or a certification that the stationary source's location does not meet either of the location criteria noted in Sections III.D.1.a.(i) or III.D.1.a.(ii).

III.F.1.c. The emissions reports required under this Section III.F.1. that estimate the actual annual emissions for calendar year 2025 emissions of formaldehyde must be submitted on a Division-approved form and in accordance with allowable procedures and exemptions set forth in Regulation Number 3, Part A, Section IX.

III.F.2. Non-Categorical Analysis Requirements

No later than October 31, 2027, the stationary source subject to this requirement pursuant to Section III.F.1.a. of this Part B must submit to the Division a one-time analysis of cost effective and technically feasible measures to reduce applicable PTAC emissions. Measures for reducing emissions may include:

III.F.2.a. Installation of control technologies.

III.F.2.b. Lower or non-emitting alternative processes.

III.F.2.c. Material substitutions.

PART C Statements of Basis, Specific Statutory Authority and Purpose

I. Adopted: January 17, 2025

This Statement of Basis, Specific Statutory Authority, and Purpose complied with the requirements of the Colorado Administrative Procedure Act § 24-4-103(4), C.R.S., the Colorado Air Pollution Prevention and Control Act §§ 25-7-110, -110.5, and -110.8 C.R.S. (the State Air Act), and the Air Quality Control Commission's (Commission) Procedural Rules, 5 Code Colo. Reg. § 1001-1.

Basis

In 2022, the General Assembly adopted House Bill (HB) 22-1244 (Public Protections from Toxic Air Contaminants), now codified primarily in § 25-7-109.5, C.R.S. § 25-7-109.5(6)(a)(I), C.R.S. directs the Commission to adopt rules that identify up to five priority toxic air contaminants (PTACs) by no later than April 30, 2025.

The Commission adopted a new Regulation Number 30 to meet the statutory directive at § 25-7-109.5(6)(a)(I), C.R.S.

Specific Statutory Authority

The State Air Act, specifically § 25-7-105(1), C.R.S., directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in § 25-7-102, C.R.S. and that are necessary for the proper implementation and administration of the State Air Act. § 25-7-109.5(6)(a)(I), C.R.S. directs the Commission to, by no later than April 30, 2025, identify up to five PTACs, considering:

- I. Existing data concerning toxic air contaminants gathered through the Air Pollution Control Division (Division) Monitoring programs;
- II. Data reported to the Division concerning emissions of toxic air pollutants;
- III. Data reported to the federal toxics release inventory pursuant to 42 U.S.C. § 11023 and data prepared by the federal environmental protection agency's (EPA) air toxics screening assessment (AirToxScreens) program;
- IV. Any other relevant data submitted to the Commission during the rule-making process concerning the amount of emissions and concentrations of toxic air contaminants in the ambient air of the state, including data collected through community-led monitoring programs; and
- V. Input from the scientific community.

Purpose

As directed in § 25-7-109.5(6)(a)(I), C.R.S., the Commission adopted a new Regulation Number 30, which identifies five PTACs.

Regulation Number 30 Regulatory Language and Outline

In Part A, General Provisions for Toxic Air Contaminants, the Commission adopted necessary definitions. In Part B, Identification of Priority Toxic Air Contaminants, Section 1, the Commission identified five PTACs.

Prioritization Framework

The Commission adopted the five PTACs based on a prioritization framework developed by Colorado Department of Public Health and Environment (CDPHE) staff. This prioritization framework involved input from a scientific community technical working group, feedback from stakeholders, and months of research and data analysis, consistent with the requirements set forth in § 25-7-109.5(6)(a)(I), C.R.S.

CDPHE staff collected data on statewide toxic air contaminants (TACs), following requirements in § 25-7-109.5(6)(a)(I), C.R.S. These data sources included existing data concerning toxic air pollutants gathered through Division monitoring programs, data reported to the Division concerning emission of toxic air pollutants, data reported to or prepared by the EPA for the Toxic Release Inventory (TRI) and EPA's AirToxScreen program, data collected through community-led monitoring programs, and input from the scientific community. These data sources were supplemented with additional monitoring and modeled data suggested by the scientific community technical working group, and data from the first annual reporting cycle of TAC that was required under § 25-7-109.5(4)(a), C.R.S.

Using these data to understand the prevalence of different TACs in Colorado, staff then estimated cancer and non-cancer health risks using pollutant-specific toxicity values. These toxicity values were adjusted slightly during the rulemaking process based on updates that EPA made to its regional screening level (RSL) table in November 2024. Toxicity values are numerical indicators of the potential of a chemical to cause harmful effects and are specific to either cancer or non-cancer health impacts. Cancer and non-cancer health impacts are analyzed separately due to differences in dose-response modeling. Chemicals that cause non-cancer impacts are generally considered to have a threshold below which there are no health effects. By contrast, even small exposures to chemicals that cause cancer are thought to increase a person's cancer risk.

Non-cancer health impacts are expressed as a Hazard Quotient (HQ). A HQ below 1 indicates that negative health effects are not expected. If the HQ exceeds 1, adverse health effects may be more likely to occur. Cancer health impacts i.e., the increased potential to develop cancer—is expressed as the increased risk of cancer per 1 million people.

For pollutants with adequate monitored or modeled data from the sources described above, the Division calculated screening level risk estimates for both cancer and non-cancer health impacts using the pollutant-specific toxicity values. This analysis showed a measure of health risk based on modeled or monitored ambient concentrations of certain TACs.

CDPHE staff then assigned groupings to TACs using a tiered health risk approach. For the purposes of prioritization, CDPHE used a cancer screening risk level of greater than or equal to 1 in 1 million excess cancer cases. For non-cancer health impacts, CDPHE used a screening risk level of a HQ of greater than or equal to 0.1. The threshold of a HQ of 0.1 is commonly used during screening when multiple chemicals are present. Though negative health impacts are not expected for a single chemical at an HQ of 0.1, it is used in screening in order to account for exposures to multiple chemicals that could act together to cause negative health impacts.

Once the pollutants meeting or exceeding the screening levels for modeled or monitored data were identified, staff considered additional factors in analyzing those pollutants. The Division compared the existing monitored or modeled data from the list of approximately 40 pollutants to the various reporting program datasets it had available (e.g., APEN, National Emissions Inventory (NEI) and TRI). The emissions reported in each dataset were weighted by a relative toxicity factor from the EPA Risk Screening Environmental Indicator (RSEI) model. These toxicity-weighted emissions were ranked from highest to lowest. All of the top 11 toxicity-weighted pollutants in each dataset were also identified in the list of 40 pollutants screened in the aforementioned risk thresholds, further supporting the potential prioritization of pollutants on that list.

For modeled risk, staff evaluated the predominant source of each TAC that fell into this category. TACs predominantly from sources that the Commission has authority to regulate were prioritized over TACs predominantly resulting from sources outside the scope of Commission rules (e.g., biogenic sources and wildfires). Separately, staff limited its assessment for modeled risk to non-mobile sources. This was done to narrow the list of potential TACs to no more than five PTACs, consistent with the statutory direction. Staff determined that excluding mobile source emissions was reasonable because § 25-7-109.5(7)(b)(II), C.R.S. specifically directs the Commissions to consider stationary source emissions in developing emission control regulations for PTACs and because § 25-7-109.5(8)(a), C.R.S. specifically directs the Division to assess its needs to administer a PTAC stationary source permitting program. Additionally, Colorado is currently pursuing several incentive and regulatory programs to reduce emissions from mobile sources, such as state enterprises created under Senate Bill 21-260, including the Clean Fleet Enterprise, Nonattainment Area Pollution Mitigation Enterprise, Clean Transit Enterprise and Community Access Enterprise, the Colorado Clean Cars Program and the Advanced Clean Trucks and Heavy-Duty Low NOx rules. These mobile source programs are anticipated to begin achieving significant reductions of TACs as a co-benefit. These modeled risks were then ranked from highest to lowest for cancer risk and non-cancer HQ.

Over three meetings with the scientific community technical working group, staff posed questions and gathered feedback from group members on the prioritization framework. Specifically, the group provided insight into quantifying risk from monitored and modeled data, how to develop a risk-based framework, use of different toxicity values, and which health effects to focus on during prioritization of TACs. Group members also supplemented additional scientific research and sources of monitoring and modeling data to be used in the prioritization framework.

Using the prioritization framework, staff identified three PTACs that had an estimated HQ over 1 for non-cancer health impacts or over 100 in 1 million cancer risk. These were acrolein, ethylene oxide, and hydrogen sulfide. Staff then considered populations exposed to TACs that ranked among the top five modeled risks for cancer or non-cancer to assess risk based on how many people may be impacted by these pollutants, including how many of those people are located in a disproportionately impacted community as prioritized in the Air Quality Control Commission (AQCC) Regulation (Reg.) 3 Disproportionately Impacted Community Layer in Version 1.0 of Colorado EnviroScreen. The Division used this layer because it was designed to prioritize areas that are relevant for stationary source regulations, and staff focused on stationary source emissions in selecting the PTAC. Using census block and block group averages from AirToxScreen 2020 data, staff estimated the number of people exposed to TACs from non-mobile, anthropogenic sources of emissions. This analysis allowed staff to identify three additional pollutants, formaldehyde, benzene, and hexavalent chromium, for prioritization, over others with similar risk characterizations, given the exposure and risk posed to more people state-wide. The Commission prioritized formaldehyde over acrolein based on this population exposure analysis and feedback and data submitted during the rulemaking process.

Identification of Five Priority Toxic Air Contaminants

Applying the foregoing analysis, the Commission adopted benzene, chromium compounds (hexavalent), ethylene oxide (EtO), formaldehyde and hydrogen sulfide (H₂S) as the initial five PTACs.

Benzene

Benzene emissions met the screening threshold of over 1 in 1 million cancer risk based on the estimated health risk from both monitoring and modeling data. Benzene is a federally designated HAP. In Colorado, based on the EPA AirToxScreen modeling results for human-caused, non-mobile sources, and considering adjustments made to correct for inventory data used in the AirToxScreen modeling, benzene was among the top five highest cancer risk based on AirToxScreen 2019 and among the top six highest cancer risk based on AirToxScreen 2020 modeling results. In addition to this pollutant ranking among the highest modeled cancer risk pollutants, staff further prioritized this pollutant based on population-level exposures, finding there are approximately 180 people, of which 157 are located in a disproportionately impacted community, exposed to benzene in Colorado at levels with an estimated risk of at least 10 excess cancer cases per million people. There are also 1.4 million people, of which 697,000 are

located in a disproportionately impacted community, exposed to benzene in Colorado at levels with an estimated risk of between 1 and 10 excess cancer cases per million people.

In addition to the modeled cancer risk, the available annual average monitoring data for over 100 different monitored benzene concentrations is in the 1-10 per million excess cancer cases range for the most part. Specifically, the available monitoring data ranges from 1 to 27 per million excess cancer cases, with a mean of approximately 6 per million excess cancer cases. Benzene is a volatile organic compound (VOC), a class of chemicals released into the air from numerous sources, including vehicle exhaust, cleaning products, paint, and gasoline. In Colorado, oil and gas operations, along with other industrial activities, are significant sources of benzene emissions. Benzene is naturally present in crude oil and natural gas, and significant emissions occur during various stages of the handling of crude oil and natural gas. During the extraction process, benzene is released when oil and gas is brought to the surface, as the volatile compounds are exposed to the atmosphere. In the refining process, crude oil is separated into different products, releasing benzene during the breakdown of hydrocarbons. Similarly, when natural gas is processed for sale, benzene may be released during various processing steps. Storage and transportation of oil and gas also contribute to benzene emissions, especially if containment systems are not properly sealed, allowing volatile compounds to escape into the air. Additionally, benzene emissions occur from combustion of wood including residential burning and forest fires. Chronic exposure to benzene may cause disorders in the blood by affecting bone marrow. Aplastic anemia (a risk factor for a specific type of leukemia), excessive bleeding, and damage to the immune system (by changes in blood levels of antibodies and loss of white blood cells) may develop. Benzene exposure may cause structural and numerical chromosomal abnormalities. Animal and human studies have provided evidence that exposure to benzene may impact reproductive and developmental organs. The EPA classifies benzene as a carcinogen. Long term exposure to benzene can cause leukemia. There have also been studies that show an association between benzene exposure and types of lymphoma, myeloma, and lung cancers.

Chromium

Chromium compounds, hexavalent, also called hexavalent chromium, had an estimated excess cancer risk that met the screening threshold of over 1 in 1 million cancer risk based on the estimated health risk from modeled data after corrections to inventory data were made. Chromium compounds are federally designated HAPs and hexavalent chromium is a state designated HAP under § 25-7-109.3(5)(a) C.R.S. In Colorado, based on the EPA AirToxScreen modeling results for human-caused, non-mobile sources, and considering adjustments made to correct for inventory data used in the AirToxScreen modeling, hexavalent chromium was among the top five highest cancer risk based on AirToxScreen 2019 and 2020. Staff further prioritized this pollutant based on population-level exposures, finding that, based on the updated RSL values and EPA AirToxScreen 2020 modeling results for human-caused, non-mobile sources, approximately 33 people, all of which are located in a disproportionately impacted community, are exposed to hexavalent chromium at levels with an estimated risk of at

least 10 excess cancer cases per million people and 7,800 people, of which 5,400 are located in a disproportionately impacted community, are exposed to hexavalent chromium at levels with an estimated risk of between 1 and 10 excess cancer cases per million people. This population exposure is the next highest after benzene among pollutants that had the top five highest modeled cancer or non-cancer health impacts, with the exception of formaldehyde.

Hexavalent chromium is generated when chromium compounds combine with other elements during industrial processes. These compounds are widely used in various manufacturing applications, such as in the production of metal alloys (i.e., stainless steel, brass, aluminum, or plastic) and in decorative chromium electroplating for car parts, furniture, hydraulic cylinders, and aircraft landing gears. Hexavalent chromium is also utilized in anodizing operations to prevent corrosion in aerospace components and their machined parts. Hexavalent chromium is released into the air during the combustion of oil, natural gas, coal, and in certain manufacturing processes, including cement and hazardous waste facilities. Cigarette smoke and industrial emissions from metal processing are additional sources. Furthermore, it is released from plating tanks through control devices and vents, windows, and doors as fugitive emissions during chrome plating operations.

Coal-fired power plants are the largest emitter in Colorado followed by fabricated metal product manufacturing, particularly processes used in electroplating operations and abrasive blasting of metal parts. The sources driving the highest risk from hexavalent chromium are these smaller sources due to their proximity to more densely populated areas. Chronic exposure to hexavalent chromium may cause respiratory effects, including perforations and ulcerations of the septum, bronchitis, decreased pulmonary function, pneumonia, asthma, and nasal itching or soreness. There is limited evidence that exposure to hexavalent chromium may result in complications during pregnancy and childbirth. Epidemiological studies of workers have found clear associations between chromium exposure and lung cancer. EPA, DHHS, and IARC have all classified hexavalent chromium as a carcinogen.

Ethylene Oxide

Ethylene oxide (EtO) emissions had an estimated cancer risk exceeding 100 in 1 million exposed people, based on the estimated health risk from both modeling data, including only human-caused, non-mobile sources, and monitoring data in the state. EtO is a federally designated HAP. EtO is a colorless gas, used in making a range of products such as antifreeze, textiles, plastics, detergents, and adhesives. EtO is also used to sterilize equipment and plastic devices, often medical equipment, unable to be cleaned by steam. In Colorado, sterilizers located in commercial sterilization facilities, or supporting other industrial processes or research and development applications, are the largest contributors to EtO emissions. The storage of sterilized materials is also one source of emissions.

While the EPA finalized a regulation on April 5, 2024, see 89 Fed. Reg. 24090, to reduce emissions from four existing commercial sterilizers as well as any new sterilizer

facility that should be built in the state, there are other sources of EtO in the state that are not covered by EPA's rule. Additionally, this PTAC had the highest estimated cancer risk of any pollutant in the state. Chronic exposure to EtO may cause eye, skin, and respiratory irritation, as well as impacts to the nervous system, including headache, nausea, memory loss, and numbness. EtO exposure may cause an increased rate of miscarriages as well as other reproductive effects. The EPA classifies EtO as a carcinogen. EtO has been shown to cause lymphoid cancer and breast cancer in humans.

Formaldehyde

Formaldehyde met the screening threshold of over 1 in 1 million cancer risk based on the estimated health risk from both monitoring and modeling data. Additionally, formaldehyde met the screening threshold of an HQ over 0.1 from both monitoring and modeling data. Formaldehyde is a federally-designated HAP. Formaldehyde is a colorless, flammable gas at room temperature with a strong, pungent odor and is highly reactive and is classified as a VOC. In Colorado, based on the EPA AirToxScreen modeling results for human-caused, non-mobile sources, and considering adjustments made to correct for inventory data used in the AirToxScreen modeling, formaldehyde was among the top five highest cancer risk based on AirToxScreen 2019 and AirToxScreen 2020 modeling results. Formaldehyde also had the sixth highest HQ for AirToxScreen 2020 modeling results.

In addition to this pollutant ranking among the highest modeled cancer risk pollutants, staff further prioritized this pollutant based on population-level exposures, finding there are approximately 400,000 people, of which 2,000 are located in a disproportionately impacted community, exposed to formaldehyde from human-caused, non-mobile sources in Colorado at levels with an estimated risk of at least 10 excess cancer cases per million people. There are also 4.15 million people, of which 93,000 are located in a disproportionately impacted community, exposed to formaldehyde in Colorado at levels with an estimated risk of between 1 and 10 excess cancer cases per million people. Some of this same population, approximately 229,000 people, of which 2,000 are located in a disproportionately impacted community, is also exposed to an HQ between 0.1 and 0.6. In addition to the modeled cancer risk, the available annual average monitoring data for 13 different monitored formaldehyde observations ranges from 15 to 57 per million excess cancer cases, with a mean of approximately 38 per million excess cancer cases. For monitored HQ, the available data is in the range of 0.17 - 0.63 with a mean of 0.41 HQ.

In Colorado, formaldehyde is a notable byproduct of incomplete combustion including the burning of organic matter, like wood (e.g., wildfires, residential wood combustion, or open burning), other combustion associated with industrial operations and electricity generation, motor vehicle exhaust, and decaying organic matter. Much of the combustion occurring in the industrial sector is associated with oil and gas extraction and production including combustion engines used for drilling or compression activities or flaring, as well as combustion to operate glycol dehydrators and regeneration units at natural gas processing plants. Formaldehyde is also emitted from surface coating or

solvent usage and during the production or of composite wood products. Secondary formation is also a significant source of formaldehyde emissions, resulting from atmospheric reaction between VOCs in the atmosphere. Chronic exposure to formaldehyde may cause respiratory symptoms, sensitization leading to asthma and contact dermatitis, neurological effects, and eye, nose, and throat irritation. Occupational and animal studies have shown epithelial lesions from chronic inhalation exposure. There is limited evidence that pulmonary functions may also be adversely affected. EPA has classified formaldehyde as a probable human carcinogen. Occupational studies have found limited evidence of associations between formaldehyde exposure and increased incidence of lung and nasopharyngeal cancer. The International Agency for Research on Cancer (IARC) and Department of Health and Human Services (DHHS) have classified formaldehyde as a human carcinogen.

Hydrogen Sulfide

Hydrogen Sulfide emissions had an estimated health risk from monitoring data that exceeded a HQ of 1 for non-cancer health impacts. Estimated health risk from modeling data, including only human-caused, non-mobile sources, met the screening criteria of a HQ over 0.1. Hydrogen Sulfide is not a HAP under federal or state law, but it is a “covered air toxic,” as defined at § 25-7-141(2)(b), C.R.S. Since Hydrogen Sulfide is not a HAP, it is not reported to the EPA National Emission Inventory (NEI) or used in EPA AirToxScreen modeling analysis. The scientific community technical working group provided feedback to expand the data sources considered to include CDPHE modeling studies to address this gap. Staff developed dispersion modeling conducted in a selected 20-kilometer area near Commerce City North Denver to assess the health risk from this pollutant. This area was selected for modeling because it has the highest concentration of reported hydrogen sulfide emissions in the state. Additionally, 3 other stationary monitors had monitoring data available for hydrogen sulfide that were above the screening criteria of a HQ of 0.1.

Hydrogen sulfide is a colorless gas released from both natural sources, such as the decay of organic matter, and human activities, including waste disposal (wastewater treatment and landfilling), industrial processes like petroleum refining, natural gas processing, wastewater treatment, roofing and asphalt material manufacturing, meat processing, gas processing plants and compressions stations, renewable natural gas processing from biogas and molybdenum mining. Chronic exposure to hydrogen sulfide may cause low blood pressure, headache, nausea, loss of appetite, weight loss, eye-membrane inflammation, loss of muscle control, chronic cough, and psychological disorders. Low concentrations can result in irritation of the eyes, nose, and throat, while higher concentrations may lead to more severe respiratory issues, headaches, dizziness, and chronic conditions.

As resources allow, the Commission encourages the Division to move forward with identifying additional PTACs sooner than 2029. In addition to other toxins the Division deems deserving of further study after consultation with scientific experts, stakeholders, and the affected public, we would encourage the Division to also study polycyclic

organic matter (including benzo(a)pyrene), ultrafine particles, and radionuclides for potential identification as a PTAC.

The Commission recognizes both the importance and complexity of this statutorily-directed rulemaking and notes that it will be followed by subsequent rulemakings to adopt health-based standards and control measures, as well as the opportunity for designating additional PTACs in the future. In selecting these five initial PTACs, the Commission relied upon the information in the record, while being responsive to community input about compounds of concern to their lived experience.

The Commission was statutorily constrained to a maximum of five toxic air contaminants in designating the initial set of PTACs. The Commission believes that the Division appropriately filtered its potential list of initial PTACs to focus first on those from stationary sources. However, there is no numeric limit in the statutes on how many TACs can be designated as PTACs throughout the life of this program. The Commission encourages the Division to not limit its study and analysis just to TACs associated with stationary sources but to also consider those associated with mobile sources to the extent supported by data.

Additional Considerations

There are no direct federal requirements that correspond with the new rule. However, to the extent the new rule could be construed to exceed or differ from any related requirements of federal law, the Commission provides the following additional statement, consistent with § 25-7-110.5(5)(a), C.R.S.:

- (I) Any federal requirements that are applicable to this situation with a commentary on those requirements;

There are no federal regulations applicable to Colorado's identification of up to five priority toxic air contaminants as such identification is a state statutory requirement. However, EPA is required under the Clean Air Act, 42 U.S.C. § 7412(b)(2), to periodically review and, where appropriate, revise the list of hazardous air pollutants established by Congress in § 7412(b)(1). EPA has, to date, identified 188 hazardous air pollutants. EPA is also required under 42 U.S.C. § 7412 to promulgate regulations establishing emission standards for each category or subcategory of major sources and area sources of hazardous air pollutants listed for regulation. Four of the five identified priority toxic air contaminants are also listed by EPA as a hazardous air pollutant.

- (II) Whether the applicable federal requirements are performance-based or technology-based and whether there is any flexibility in those requirements, and if not, why not;

The federal requirement to list hazardous air pollutants is not performance-based or technology-based. EPA adds pollutants which present, or may present, through inhalation or other routes of exposure, a threat of adverse human health or environmental effects.

- (III) Whether the applicable federal requirements specifically address the issues that are of concern to Colorado and whether data or information that would reasonably reflect Colorado's concern and situation was considered in the federal process that established the federal requirements;

The federal requirement to list hazardous air pollutants do not specifically address Colorado's statutory directive to identify up to five priority toxic air contaminants.

- (IV) Whether the proposed requirement will improve the ability of the regulated community to comply in a more cost-effective way by clarifying confusing or potentially conflicting requirements (within or cross-media), increasing certainty, or preventing or reducing the need for costly retrofit to meet more stringent requirements later;

The identification of five priority toxic air contaminants does not directly establish additional requirements on the regulated community.

- (V) Whether there is a timing issue which might justify changing the time frame for implementation of federal requirements;

This is a state-specific rule that is not implementing federal requirements. Thus, no timing issue exists.

- (VI) Whether the proposed requirement will assist in establishing and maintaining a reasonable margin for accommodation of uncertainty and future growth;

The identification of five priority toxic air contaminants does not directly establish additional requirements on the regulated community and, therefore, does not impact future growth.

- (VII) Whether the proposed requirement establishes or maintains reasonable equity in the requirements for various sources;

The identification of five priority toxic air contaminants does not directly establish additional requirements on the regulated community.

- (VIII) Whether others would face increased costs if a more stringent rule is not enacted;

The identification of five priority toxic air contaminants does not directly establish additional requirements on the regulated community.

- (IX) Whether the proposed requirement includes procedural, reporting, or monitoring requirements that are different from applicable federal requirements and, if so, why and what the "compelling reason" is for different procedural, reporting, or monitoring requirements;

The identification of five priority toxic air contaminants does not directly establish additional procedural, reporting, or monitoring requirements.

- (X) Whether demonstrated technology is available to comply with the proposed requirement;

The identification of five priority toxic air contaminants does not directly establish additional requirements on the regulated community.

- (XI) Whether the proposed requirement will contribute to the prevention of pollution or address a potential problem and represent a more cost-effective environmental gain;

The identification of five priority toxic air contaminants does not directly establish additional requirements on the regulated community.

- (XII) Whether an alternative rule, including a no-action alternative, would address the required standard.

The identification of five priority toxic air contaminants achieves the statutorily mandated directive to identify such air contaminants. An alternative list of up to five priority toxic air contaminants could achieve that statutory directive, however, the five proposed were selected considering state and federal reported toxic air pollutants data; monitoring data; input from the scientific community; and a prioritization framework that considered, among other things, the potential for emissions reductions and exposures on vulnerable populations.

Findings of Fact

§ 25-7-110.8, C.R.S., requirements do not apply to this rulemaking because the adopted rule is not directly intended to reduce air pollution. To the extent § 25-7-110.8, C.R.S. does apply to this rulemaking, after considering all the information in the record, the Commission hereby makes the determination that:

- (I) The rule is based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.
- (II) The rule is not intended, directly, to address a reduction in air pollution, or, in the alternative, is administrative in nature.
- (III) Evidence in the record supports the finding that the rule provides benefits that justify the costs to government to implement the rule, and the rule does not result in any cost to the regulated community or public.
- (IV) The rule has no regulatory impact on any person, facility, or activity and, therefore, the analysis under § 25-7-110.5(4), C.R.S. is not required. Additionally, the rule is not intended, directly, to achieve a reduction in air pollution.

(V) The rule does not directly result in air quality benefits.

APPENDICES, BY PART

Part B:

Appendix A: Priority Toxic Air Contaminants

CAS Number	Toxic Air Contaminant	Date Identified
71432	Benzene	1/17/2025
18540299	Chromium Compounds, Hexavalent	1/17/2025
75218	Ethylene Oxide	1/17/2025
50000	Formaldehyde	1/17/2025
7783064	Hydrogen Sulfide	1/17/2025

II. Adopted: September 19, 2025

This Statement of Basis, Specific Statutory Authority, and Purpose complied with the requirements of the Colorado Administrative Procedure Act section 24-4-103(4), C.R.S., the Colorado Air Pollution Prevention and Control Act sections 25-7-110, -110.5, and -110.8 C.R.S. (the State Air Act), and the Air Quality Control Commission's (Commission) Procedural Rules, 5 Code Colo. Reg. section 1001-1.

Basis

On June 2, 2022, House Bill (HB) 22-1244: Public Protections for Toxic Air Contaminants was signed into law. The law, now primarily codified at section 25-7-109.5, C.R.S., directed the Commission to propose health-based standards for each identified Priority Toxic Air Contaminant (PTAC) by April 30, 2026, for General Assembly approval. On January 17, 2025, the Commission adopted Regulation Number 30 (Regulation 30) and identified five initial PTACs. In this action, the Commission adopted revisions to Regulation 30 to propose chronic health-protective benchmarks, which meet the statutory definition of health-based standards, for each identified PTAC for General Assembly approval. Further, the Commission adopted revisions to address typographical, grammatical, and formatting errors found throughout the regulation.

The Commission did not define use-cases for the chronic health-protective benchmarks at the time of rule adoption. Without limiting the Division's existing authority under the State Air Act or Commission regulations, the Commission acknowledges that before the Division may use the chronic health-protective benchmarks as a level that could trigger a regulatory or economic impact, or as the determining factor that could trigger a regulatory or economic impact, the Commission must have promulgated an enabling regulation.

Specific Statutory Authority

The State Air Act, specifically section 25-7-105(1), C.R.S., directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in section 25-7-102, C.R.S., and that are necessary for the proper implementation and administration of the State Air Act. Section 25-7-109.5(6)(a), C.R.S. required the Commission to adopt rules that, no later than April 30, 2026, propose health-based standards for each identified PTAC for approval by the General Assembly.

Section 25-7-109.5(1)(e), C.R.S. defines health-based standards as “the chronic exposure limits for each priority toxic air contaminant required to protect the public from adverse health effects of that priority toxic air contaminant, allowing for an ample margin of safety, represented as benchmark numerical concentrations in the ambient air.”

“Adverse health effects” is defined at section 25-7-109.5(1)(a), C.R.S. as “the detrimental health effects from exposure to emissions of a toxic air contaminant, including the cumulative effects to health from exposure to the combined air emissions of the toxic air contaminant from multiple sources, whether the emissions are emitted routinely, intermittently, or accidentally.”

In determining the health-based standards, the statute required:

- I. Consideration of the best available peer-reviewed toxicity values regarding the levels of exposure to PTACs that may cause or contribute to adverse health effects;
- II. Consideration of standards adopted in other states to reduce or limit concentrations of toxic air contaminants (TACs) in the ambient air;
- III. Consideration of the effects of exposure to PTACs on vulnerable groups of the state, including disproportionately impacted communities, infants, children, fetuses, the elderly, and people with disabilities;
- IV. Consideration of both cancer-related health risks and non-cancer-related health risks and identification of the excess cancer and non-cancer risk levels used in determining the health-based standards;
- V. Provision of a sufficient margin of safety that accounts for the various effects that different populations may experience from exposure to PTACs; and
- VI. Consultation with the scientific community through holding at least one public hearing.

Purpose

Chronic Health-Protective Benchmarks

In this Regulation Number 30, the Commission adopted proposed chronic health-protective benchmarks for each PTAC to satisfy the statutory directive in section 25-7-

109.5(6)(b), C.R.S. to adopt proposed health-based standards, as informed by the factors identified in section 25-7-109.5(6)(b), C.R.S. As levels below which there is a high degree of scientific confidence that adverse health effects are unlikely to occur from exposure to each PTAC on an individual basis, the chronic health-protective benchmarks adopted by the Commission satisfy the statutory direction to adopt chronic exposure limits for each PTAC, represented as benchmark numerical concentrations in the ambient air, that protect the public from adverse health effects of that PTAC, allowing for an ample margin of safety.

The Commission included the defined term “chronic health-protective benchmark” in Regulation Number 30 for two reasons.

First, the term “chronic health-protective benchmarks” accurately describes the proposed values the Commission adopted. As explained above, the adopted values are set at levels the Commission determined to be health-protective for each individual PTAC, with an ample margin of safety. As such, the adopted values meet the statutory definition of health-based standards when that definition is read as a whole and together with the considerations listed under section 25-7-109.5(6)(b), C.R.S. on an individualized PTAC basis. Party comments during the rulemaking raised the potential for confusion around parallels between the National Ambient Air Quality Standards (NAAQS) and chronic health-protective benchmarks due to similar, though not identical, language in the federal Clean Air Act and HB 22-1244. Based on Division research for how ambient air concentrations for air toxics were developed for other state programs, the Commission determined that the chronic health-protective benchmarks are not intended to function as ambient air quality standards similar to the NAAQS. To avoid future conflation of the chronic health-protective benchmarks and ambient air quality standards, the Commission described the adopted values as chronic health-protective benchmarks, clarifying that the values represent levels below which adverse health effects are unlikely to occur, not levels that must be achieved to ensure public health is protected.

Second, adopting a definition of chronic health-protective benchmarks will allow for distinguishing the adopted values in this action from future iterations of health-based standards. Section 25-7-109.5(6)(c), C.R.S. provides that the Commission must consider in the future whether to include acute exposure limits for PTACs in the definition of health-based standards. By identifying the proposed values adopted in this action as chronic health-protective benchmarks, the Commission can easily distinguish such values from any future acute values.

The Commission also adopted revisions at Part A, Section I.A. to establish definitions essential to understanding how the chronic health-protective benchmarks were calculated. In Part B, Section II.A. the Commission adopted proposed chronic health-protective benchmarks, as set out in Appendix B. Part B, Section II.A. and Appendix B may be revised with future iterations of PTAC identification and health-based standard adoption, as required under section 25-7-109.5(6)(d), C.R.S. They may also be revised in the future to modify existing health-based standards, consistent with section 25-7-109.5(6)(c), C.R.S.

The chronic health-protective benchmark table in Part B, Appendix B, differentiates between cancer risk and/or non-cancer health risk. The table also denotes where the toxicology values were derived and the most recent year the values have been updated.

The proposed chronic health-protective benchmarks were calculated using (1) pollutant-specific toxicity values and (2) selected cancer and non-cancer health risk levels. These two inputs are discussed below.

After considering all the information in the record, the Commission selected the risk levels used in the chronic health-protective benchmarks as the appropriate levels for the state of Colorado as a matter of policy. The selected levels were informed by the statutory definitions of “health-based standards” and “adverse health effects,” and by the considerations at section 25-7-109.5(6)(b). The Commission determined the toxicity values used in the chronic health-protective benchmarks were the best available peer-reviewed toxicity values, see section 25-7-109.5(6)(b)(I), considering evidence in the record and largely in accordance with longstanding CDPHE policy.

Toxicity Values

Toxicity values are numerical values that indicate the relative toxicity of individual chemicals and their capacity to cause adverse health effects. Pollutants can have toxicity values for either or both cancer and non-cancer health risk. Toxicity values for carcinogenic pollutants are expressed as an inhalation unit risk (IUR), as defined in Section I.A.3, while toxicity values for pollutants with non-cancer health impacts are expressed as a reference concentration (RfC), as defined in Section I.A.5.

Consideration of the best available toxicity values (section 25-7-109.5(6)(b)(I), C.R.S.)

The best available toxicity values used to determine the proposed chronic health-protective benchmarks were selected largely based on a Colorado Department of Public Health & Environment (CDPHE) policy entitled *On Use Of Human Health Toxicity Values In Environmental Risk Assessment And Remediation Management*, published in 2004. The policy establishes a hierarchy of federal and some state sources to consult when deciding upon a toxicity value to use for various risk-management or policy-related decision making. This policy aligns with EPA’s approach in other programs including the Office of Air Quality Planning and Standards, AirToxScreen, and the Risk-Screening Environmental Indicators Model.

The Commission recognizes the value in following a systematic approach to the selection of toxicity values. For the non-cancer toxicity value for benzene, however, the Commission determined to use the value from the California Office of Environmental Health Hazard Assessment rather than the value from the EPA Integrated Risk Information System (IRIS), as would have been dictated by existing CDPHE policy. The Commission made this decision based on evidence presented by parties to the rulemaking and the Commission’s informed judgment. The Commission may revisit toxicity values, including the selection process, in future rulemakings to adopt new chronic health-protective benchmarks or to revise existing chronic health-protective

benchmarks, as appropriate. Consideration of margin of safety and vulnerable groups in toxicity values (sections 25-7-109.5(1)(e) and 25-7-109.5(6)(b)(III), (V), C.R.S.)

The proposed chronic health-protective benchmarks, by definition, had to allow for “an ample margin of safety.” Further, in determining the chronic health-protective benchmarks, the Commission had to consider the effects of exposure to PTACs on vulnerable groups in the state and provide a sufficient margin of safety that accounts for the various effects that different populations may experience from PTAC exposure. The selected toxicity values satisfied these requirements because all of the sources of toxicity values chosen to calculate the proposed chronic health-protective benchmarks incorporate uncertainty factors to account for variations in susceptibility within the human population. These uncertainty factors are used by different sources, like the EPA Integrated Risk Information System (IRIS), when translating scientific studies, which often only include a single group (for example a group of workers), into values that are protective for the larger public. The application of these uncertainty factors means that the toxicity values selected include considerations for differences in sensitivity between humans that may arise from differences like age and health status. The toxicity values used are all chronic toxicity values, which are developed to be protective for exposures which may occur over a person’s lifetime.

Consultation with the scientific community on toxicity values (section 25-7-109.5(6)(b)(VI), C.R.S.)

In meetings with a scientific community technical working group, staff presented options for toxicity values for the identified PTACs. Several members of the working group were supportive of using the CDPHE policy for toxicity values. Further, as required by section 25-7-109.5(6)(b)(VI), C.R.S., the Commission convened a public hearing on February 21, 2025, during which the Commission heard from the scientific community technical working group and the Division regarding the proposed framework to calculate chronic health-protective benchmarks. This briefing provided the Commission with an opportunity to receive technical perspectives and better understand the context surrounding the toxicity values being considered for the proposed chronic health-protective benchmarks. Working group members noted that the CDPHE policy was consistent with other EPA policies on toxicity value hierarchies. Some members expressed concern that newer values should be prioritized regardless of the tier of the source in the CDPHE policy.

Consideration of toxicity values used in other states (section 25-7-109.5(6)(b)(II), C.R.S.)

CDPHE staff conducted research on numerous other state regulatory agency programs for air toxics. This research included how ambient air concentrations were derived and applied across the different programs. A majority of the programs used similar sources of toxicity values as the CDPHE policy, with the most common source being EPA’s IRIS (tier 1 in CDPHE’s policy). A few states derive their own toxicity values either informed by existing sources or developed with help from scientific boards or committees. There are several states that have opted to use occupational values from agencies such as

The Occupational Safety and Health Administration, The National Institute for Occupational Safety and Health, or American Conference of Governmental Industrial Hygienists. Since these values are developed for healthy workers exposed for a typical work-span, these states may make adjustments to these values when they are used for a population level risk assessment or screening.

Risk levels

In addition to toxicity values, the other key input in determining the proposed chronic health-protective benchmarks was risk level. The statute required consideration of both cancer and non-cancer health impacts and associated risk levels when determining the proposed chronic health-protective benchmarks. Cancer risk, for the purposes of risk assessment, is often expressed as the excess cancer cases per 1 million people exposed to a given pollutant. Non-cancer health impacts are expressed using a HQ, which shows the ratio between potential exposure to a pollutant and the level at which no adverse health effects are expected from a single pollutant.

By definition, the proposed chronic health-protective benchmarks are required to be set at a level to protect the public from adverse health effects, allowing for an ample margin of safety. Further, in adopting the proposed chronic health-protective benchmarks, the Commission had to consider the effects of exposures on vulnerable groups of the state (section 25-7-109.5(6)(b)(III)) and provide a sufficient margin of safety that accounts for the various effects that different populations may experience from PTAC exposures (section 25-7-109.5(6)(b)(V)). In alignment with these directives, the Commission used a cancer risk level of 1 excess cancer case in 1 million exposed people and a non-cancer HQ of 1.0 to calculate the proposed chronic health-protective benchmarks.

A cancer risk level of 1 in 1 million aligns with long-standing precedent in federal environmental policy. In particular, EPA identifies 1 in 1 million as its target to protect the greatest number of people in its residual risk program and uses it as a screening threshold in programs like EPA Superfund when assessing risk from multiple pollutants. The cancer risk level of 1 in 1 million provides an ample margin of safety when considering vulnerable populations exposed to multiple TAC as it allows for the summing of multiple cancer risks while staying below the 100 in 1 million level, which EPA considers unacceptable.

An HQ of 1 represents the threshold level at which no adverse health effects are expected from exposure to that level of a single pollutant. A level below an HQ of 1 is common practice in federal and state risk assessments where multiple pollutants and shared target organs are involved.

Consideration of adverse health effects, margin of safety, and vulnerable groups in risk levels (sections 25-7-109.5(1)(a), 25-7-109.5(1)(e), 25-7-109.5(6)(b)(III), (V), C.R.S.)

The risk levels selected by the Commission to calculate the proposed chronic health-protective benchmarks considered that the standards must provide an ample margin of safety and consider vulnerable groups. The Commission recognizes that Coloradans

are not exposed to emissions from a single pollutant at a time, and that the additive effects from multiple exposures may exacerbate risk. Because the toxicological data that are the basis of the CHPB are typically collected for only a single pollutant at a time, multi-pollutant impacts are not included in these CHPB values. This important consideration will be addressed in future rulemakings on this topic.

Consultation with the scientific community on risk levels (section 25-7-109.5(6)(b)(VI), C.R.S.)

In meetings with the scientific community technical working group, staff presented options for risk levels for the identified PTACs. The scientific community working group did not provide a consensus recommendation on the cancer or non-cancer risk levels to be used in the chronic health-protective benchmarks. Some conveyed interest in using the most conservative values, considering that people do not breathe a single PTAC in isolation, while others were concerned that they could not advise on the risk levels without knowing how the resulting chronic health-protective benchmarks might be used. This feedback was shared with the Commission during a public hearing on February 21, 2025, with the Commission.

Consideration of risk levels adopted in other states (section 25-7-109.5(6)(b)(II), C.R.S.)

Staff also considered risk levels used in other states. Based on CDPHE staff research, the most commonly chosen cancer risk level in other state programs was 1 in 1 million excess cancer cases. Maine, Minnesota, Texas, and Maryland were in the minority by choosing 10 in 1 million excess cancer cases. Importantly, Minnesota uses the same cancer risk level, of 10 in 1 million excess cancer cases, for an individual carcinogen and for the sum of multiple carcinogens from a source.

For non-cancer risk levels, the majority of state programs calculate their benchmark values using an HQ of 1. Massachusetts and Texas both use values that are lower than an HQ of 1. The values that the Texas uses in their permitting program are set at a level that is effectively an HQ of 0.3 in order to “account for exposure to chemicals from multiple sources.” Massachusetts uses an HQ of 0.2.

Calculations for Proposed Chronic Health-Protective Benchmarks

Considering the above factors, staff considered the best available peer-reviewed information concerning PTAC toxicity values, collaborated internally across departments, and sought input from a variety of stakeholders, including the scientific community, to select the appropriate level of risk to calculate the proposed chronic health-protective benchmarks. The risk levels adopted by the Commission reflect statutory directives to propose chronic health-protective benchmarks that consider adverse health effects and incorporate an ample margin of safety, including for vulnerable groups.

To calculate the proposed cancer chronic health-protective benchmarks, which are presented as a concentration in the ambient air, micrograms per meter ($\mu\text{g}/\text{m}^3$), standard methodologies were used to convert the toxicity values and risk levels into

concentrations. The cancer risk level (0.000001, representing 1 excess cancer case per 1 million people) is divided by the IUR, for each PTAC. For non-cancer risk, the RfC is multiplied by the risk level (HQ of 1.0) for each PTAC and then multiplied by 1,000 to align the units of the RfC with the desired units of each proposed chronic health-protective benchmark ($\mu\text{g}/\text{m}^3$).

Scope of the Chronic Health-Protective Benchmarks

Although the statute defines a health-based standard and sets forth statutory requirements for the development of health-based standards, it does not prescribe specific regulatory applications of the proposed health-based standards. In assessing how the proposed chronic health-protective benchmarks could be applied in the future, staff research concluded that most states use benchmark concentrations to compare to modeled emissions estimates from potential or existing sources during the permitting process in order to determine if additional information or regulatory action might be required, or if the permit application might require refinements to reduce the risk of exposure to one or more air toxic pollutants. Some states also use benchmark concentrations to evaluate monitoring data to identify potential areas of health concern for further study. There are also several examples where the benchmark concentrations are used as a requirement that permitted facilities must meet, as measured at the facilities' fencelines.

While the Commission is not adopting use-cases for the proposed chronic health-protective benchmarks at this time, these benchmark numerical concentrations will inform the recommended permitting framework that will be published in the permitting needs assessment submitted to the General Assembly in December 2025. The Commission also understands that staff will present the proposed chronic health-protective benchmarks for approval at this SMART Act hearing in January 2026 in order to maximize the time available during the 2026 session for the General Assembly to take action on the proposed values and provide any direction for a potential future permitting program.

The Commission recognizes that section 25-7-109.5(6)(c), C.R.S. allows for an iterative review of the chronic health-protective benchmarks, including incorporation of updated toxicity values and/or risk levels over time. As part of this iterative review, the Commission may determine whether to include acute exposure limits for PTACs in this Regulation Number 30.

Additional Considerations

There are no direct federal requirements that correspond with the new rule. However, to the extent the new rule could be construed to exceed or differ from any related requirements of federal law, the Commission provides the following additional statement, consistent with section 25-7-110.5(5)(a), C.R.S.:

- (l) Any federal requirements that are applicable to this situation with a commentary on those requirements;

There are no federal requirements applicable to Colorado's proposal for chronic health-protective benchmarks for each identified PTAC, as such identification is a state statutory requirement. While the federal framework for the National Emissions Standards for HAPs (NESHAPs) include risk assessment for certain source categories, there is no direct analog to the proposed chronic health-protective benchmarks required under section 25-7-109.5(6)(a)(II), C.R.S.

- (II) Whether the applicable federal requirements are performance-based or technology-based and whether there is any flexibility in those requirements, and if not, why not;

Most NESHAPs are technology-based. However, the promulgation of the proposed chronic health-protective benchmarks is separate from and unrelated to these federal standards.

- (III) Whether the applicable federal requirements specifically address the issues that are of concern to Colorado and whether data or information that would reasonably reflect Colorado's concern and situation was considered in the federal process that established the federal requirements;

Although the federal NESHAP framework includes residual health risk assessment for certain types of source categories, and health risk was also considered in proposing the chronic health-protective benchmarks, the NESHAP framework does not specifically address Colorado's statutory directive to propose chronic health-protective benchmarks for the identified PTACs. The statutory directive to propose such chronic health-protective benchmarks is separate from and does not mirror the source category-based NESHAP framework. Additionally, NESHAPs only apply to federally-designated HAPs, whereas the list of PTACs—for which chronic health-protective benchmarks must be proposed—includes pollutants that are not federally-designated HAPs.

- (IV) Whether the proposed requirement will improve the ability of the regulated community to comply in a more cost-effective way by clarifying confusing or potentially conflicting requirements (within or cross-media), increasing certainty, or preventing or reducing the need for costly retrofit to meet more stringent requirements later;

The proposed chronic health-protective benchmarks have no specified regulatory application and do not establish additional requirements on the regulated community.

- (V) Whether there is a timing issue which might justify changing the time frame for implementation of federal requirements;

This is a state-specific rule that is not implementing federal requirements. Thus, no timing issue exists.

- (VI) Whether the proposed requirement will assist in establishing and maintaining a reasonable margin for accommodation of uncertainty and future growth;

The proposed chronic health-protective benchmarks for the five PTACs do not establish additional requirements on the regulated community and, therefore, do not impact future growth.

- (VII) Whether the proposed requirement establishes or maintains reasonable equity in the requirements for various sources;

The proposed chronic health-protective benchmarks do not establish additional requirements on the regulated community.

- (VIII) Whether others would face increased costs if a more stringent rule is not enacted;

There are no federal requirements applicable to Colorado's proposal for chronic health-protective benchmarks and the proposed chronic health-protective benchmarks have no specified regulation application. As a result, whether others would face increased costs if a more stringent rule is not enacted is not relevant to this rulemaking.

- (IX) Whether the proposed requirement includes procedural, reporting, or monitoring requirements that are different from applicable federal requirements and, if so, why and what the "compelling reason" is for different procedural, reporting, or monitoring requirements;

The proposed chronic health-protective benchmarks do not establish additional procedural, reporting, or monitoring requirements.

- (X) Whether demonstrated technology is available to comply with the proposed requirement;

The proposed chronic health-protective benchmarks do not establish additional requirements on the regulated community.

- (XI) Whether the proposed requirement will contribute to the prevention of pollution or address a potential problem and represent a more cost-effective environmental gain;

The proposed chronic health-protective benchmarks do not establish additional requirements on the regulated community.

- (XII) Whether an alternative rule, including a no-action alternative, would address the required standard.

The proposed chronic health-protective benchmarks achieve the statutorily mandated directive to propose health-based standards. A no-action alternative

would not meet the statutory directive. Alternative levels for the chronic health-protective benchmarks might exist, but would have to meet the statutory requirements, including that they must be set at levels to protect the public from adverse health effects (defined as including cumulative effects from multiple sources), that they must allow an ample margin of safety, and that they are otherwise supported by the criteria set out in section 25-7-109.5(6)(b), C.R.S. The chronic health-protective benchmarks proposed were selected to meet these requirements and considering the best available peer-reviewed toxicity values; both cancer-related and non-cancer-related health risks; standards adopted in other states; the effects of exposure on vulnerable groups of the state; a sufficient margin of safety; and input from the scientific community.

Findings of Fact

Section 25-7-110.8, C.R.S., requirements do not apply to this rulemaking because the adopted rule is not directly intended to reduce air pollution. To the extent section 25-7-110.8, C.R.S. does apply to this rulemaking, after considering all the information in the record, the Commission hereby makes the determination that:

- (I) The rule is based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.
- (II) The rule is not intended, directly, to address a reduction in air pollution, or, in the alternative, is administrative in nature.
- (III) Evidence in the record supports the finding that the rule provides benefits that justify the costs to government to implement the rule, and the rule does not result in any cost to the regulated community or public.
- (IV) The rule has no regulatory impact on any person, facility, or activity and, therefore, the analysis under section 25-7-110.5(4), C.R.S. is not required. Additionally, the rule is not intended, directly, to achieve a reduction in air pollution.
- (V) The rule does not directly result in air quality benefits.

Appendix B: Chronic Health-Protective Benchmarks for Priority Toxic Air Contaminants

Table Notes:

1. Each chronic health-protective benchmark is in units of micrograms per cubic meter ($\mu\text{g}/\text{m}^3$).
2. The cancer chronic health-protective benchmark is based on a risk level of 1 excess cancer cases per million people exposed. Selected toxicity value reference includes the age dependent adjustment factor-adjustment for compounds classified as mutagens (i.e., ethylene oxide, hexavalent chromium, formaldehyde).
3. The non-cancer chronic health-protective benchmark is based on a risk of HQ equal to 1.0.

		Proposed Chronic Health-Protective Benchmarks (µg/m³) as of September 19, 2025.					
		Chronic Cancer ²			Chronic Non-Cancer ³		
		Cancer Chronic Health- Protective Benchmark (µg/m³)	Toxicity Value Reference		Non- Cancer Chronic Health- Protective Benchmark (µg/m³)	Toxicity Value Reference	
CAS Number	Priority Toxic Air Contaminant		Source	Year		Source	Year
71432	Benzene	0.13	IRIS	2000	3	CalEPA	2014
18540299	Chromium Compounds, Hexavalent	0.000056	IRIS	2024	0.03	IRIS	2024
75218	Ethylene Oxide	0.0002	IRIS	2016	30	CalEPA	2000
50000	Formaldehyde	0.091	IRIS	2024	7	IRIS	2024
7783064	Hydrogen Sulfide				2	IRIS	2003

III. Adopted: April 17, 2026

This Statement of Basis, Specific Statutory Authority, and Purpose complied with the requirements of the Colorado Administrative Procedure Act section 24-4-103(4), C.R.S., the Colorado Air Pollution Prevention and Control Act sections 25-7-110, -110.5, and -110.8 C.R.S. (the State Air Act), and the Air Quality Control Commission's (Commission) Procedural Rules, 5 Code Colo. Reg. section 1001-1.

Basis

On June 2, 2022, House Bill (HB) 22-1244: Public Protections for Toxic Air Contaminants was signed into law. The law, now primarily codified at Section 25-7-109.5, C.R.S., directed the Commission to adopt emission control regulations for each identified Priority Toxic Air Contaminant (PTAC) by April 30, 2026. On January 17, 2025, the Commission adopted Regulation Number 30 (Regulation 30) and identified five initial PTACs. In this action, the Commission adopted revisions to Regulation 30 and Regulation Number 24 to establish emission control regulations for certain stationary sources that emit one or more PTACs. Additional requirements were adopted for additional reporting of PTAC emissions and to require site-specific analyses from certain sources not subject to the emission control regulations. The emission control regulations will reduce PTAC emissions, focusing on specific source categories and the locations of sources that are in proximity to certain locations including proximity to receptors within PTAC cancer risk areas or proximity to occupied areas that are in or near disproportionately impacted communities areas for this initial set of strategies.

Section 25-7-109.5(7)(d), C.R.S. requires the Commission to revisit existing PTAC emission control regulations and adopt emission control regulations for any additional pollutant identified as a PTAC at least every five years. Consistent with this direction, the Commission may revise these initial regulations and relevant applicability criteria, including by adopting additional strategies for the initial PTACs and future PTACs, in subsequent rulemaking actions in order to continue to target further emissions reductions.

Specific Statutory Authority

The State Air Act, specifically Section 25-7-105(1), C.R.S., directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in Section 25-7-102, C.R.S., and that are necessary for the proper implementation and administration of the State Air Act. Section 25-7-109.5(7)(a), C.R.S. required the Commission to, no later than April 30, 2026, adopt emission control regulations to reduce emissions of each PTAC and prioritize reduction in disproportionately impacted communities with multiple sources of emissions of PTACs.

In determining the emission control regulations, Section 25-7-109.5(7)(b), C.R.S. requires that the Commission consider:

- I. Any emission control regulations adopted for PTACs in other states or by the federal government;
- II. The emission levels of a PTAC from different industries and categories of sources, including sources required to have an operating permit pursuant to Section 25-7-114.3, synthetic minor sources, and minor sources;
- III. The degree of reduction of each PTAC that is achievable and technically and economically feasible, taking into account energy, environmental, and economic impacts and other costs pursuant to the requirements described in Section 25-7-110.8;
- IV. The ability of emission control regulations to reduce or eliminate the emissions of a PTAC, including non-emitting alternative processes and control technologies; and
- V. The availability, suitability, and relative efficacy of a less hazardous substitute for a PTAC.

In addition, for new PTAC emission sources, Section 25-7-109.5(7)(c), C.R.S. directs the Commission to adopt emission control regulations that are more stringent than those adopted for existing sources. It also provides that the Commission may adopt an emission threshold below which new sources are not required to comply with the more stringent regulations.

“Emission control regulation” is broadly defined at Section 25-7-103(11), C.R.S. Emission control regulations pertaining to PTACs “may include application of measures, processes, methods, systems, or techniques, including, but not limited to, measures that: (a) [r]educe the volume of, or eliminate emissions of, such pollutants through process changes, emission limitations, control technologies, substitution of materials, or other modifications; (b) [e]nclose systems or processes to eliminate emissions; (c) [c]ollect, capture, or treat such pollutants when released from a process, stack, storage, or fugitive emissions points; (d) [a]re design, equipment, or work practice standards (including requirements for operator training or certification); or (e) [a]re a combination of the provisions of paragraphs (a) to (d)”

In addition to Section 25-7-109.5, C.R.S., Section 25-7-109, C.R.S. provides the Commission authority to adopt emission control regulations for PTACs and Section 25-7-106(6), C.R.S. provides the Commission authority to require sources to establish and maintain reports; install, use, and maintain monitoring equipment or methods; record, monitor, and sample emissions; and provide such other information as the Commission may require.

Purpose

The following section sets forth the Commission’s purpose in adopting revisions to Regulation Number 24 and Number 30. As directed in Section 25-7-109.5(7), C.R.S., the Commission adopted emission control regulations for each PTAC.

PTAC Emissions Control Regulations

The Commission adopted the emission control regulations for the five initial PTACs based on research and stakeholder outreach conducted by Division staff. Staff developed the emission control regulations following the direction at Section 25-7-109.5(7)(b), C.R.S. Staff compiled and analyzed emissions data from stationary sources and the processes that result in emissions. Staff researched emission control regulations for the five PTAC as adopted in other states and by the federal government, emission levels from different types of source categories, considered the degree of PTAC emissions reductions achievable and technically and economically feasible, considered the ability of emission control regulations to reduce or eliminate PTAC emissions, and considered the availability, suitability, and efficacy of less hazardous substitutes. Staff also held informational meetings with the public and industry groups, as well as source-specific stakeholder meetings throughout the rule development process to ensure adopted provisions effectively reduce PTAC emissions in ways that are technically and economically feasible.

Progress Updates and Reports to the Public

CRS 25-7-109.5(7) directs the Commission to "adopt emission control regulations to reduce emissions of each priority toxic air contaminant and prioritize reductions in disproportionately impacted communities with multiple sources of emissions of priority toxic air contaminants." Recognizing the communities' stake in this program, the Commission encourages Division staff to report progress under the rule in a user-friendly dashboard format, consistent with the current Division efforts that the Commission understands are underway. In its annual reports to the public, the Commission will include summary information on the progress of the PTAC program including available monitoring data and emissions trends.

Applicability Criteria

The Commission adopted applicability criteria that vary for sources of formaldehyde and hydrogen sulfide in order to allocate finite resources and prioritize the types of new, modified, and existing sources that must comply with these initial formaldehyde and hydrogen sulfide emission control regulations. Applicability criteria fall into two categories: emission thresholds and location-based criteria.

Emission Threshold

The primary applicability criteria for formaldehyde and hydrogen sulfide is a stationary source-wide annual emission threshold. The categories of source types and equipment and activities that emit these PTAC vary widely, and thresholds are one way to prioritize the initial number of control regulations and associated emissions reductions. The Commission did not adopt an emission threshold for ethylene oxide or hexavalent chromium because the applicability for these two pollutants was focused on sources with specific processes that were known to emit these PTAC in Colorado. While the Division's staff analysis documented which source categories emitted different amounts

of benzene, the Commission did not adopt an emission threshold for benzene emission control regulations because it adopted a control requirement for one source category – petroleum refineries. There is only a single source belonging to this category in the state of Colorado at this time and, therefore, further prioritization was unnecessary.

The adopted emission thresholds were informed by long-standing thresholds used in New Jersey, which were adopted in 1998 and derived from de minimis emission levels proposed by EPA to determine case-by-case maximum achievable control technology standards for federal HAP. The Commission established the formaldehyde PTAC applicability emission threshold as 4,000 pounds per year (2 tons per year), which is below the federal thresholds for determining a major source of hazardous air pollutants (HAP) (10 tons per year or more of a single HAP or 25 tons per year or more of combined HAP emissions). Establishing applicability thresholds below major source HAP thresholds will advance the state of technically and economically feasible PTAC control strategies by addressing sources of PTAC emissions in Colorado that are not addressed under a corresponding federal program. For hydrogen sulfide, the only PTAC that is not a federal HAP, the Commission adopted a threshold of 5 tons per year, which is consistent with the threshold rate for any toxic substance in New Jersey that is not classified as a HAP. Similar to Colorado, New Jersey recently added hydrogen sulfide as a state-only HAP. The Commission recommends the Division review whether alternative emission thresholds are appropriate during development of the next iteration of PTAC emission control regulations, considering available resources, in order to refine the types of control strategies available and appropriate for Colorado.

To determine which existing sources have emissions above the thresholds for formaldehyde and hydrogen sulfide, the Division reviewed all available data across several datasets. These included the first two years of annual TAC reports required under Section 25-7-109.5(4)(a), C.R.S., and EPA's National Emission Inventory (NEI) data from 2020. This data was further supplemented with the 2025 APEN inventory data to fill in data gaps. For hydrogen sulfide, the Division also reviewed the most recently reported EPA Toxic Release Inventory (TRI) from 2024, along with 2025 APEN data and two years of annual TAC reports. These data sources helped the Commission identify a priority set of emission control regulations focused on the key emitting equipment and activities at sources above the emission threshold.

For the purposes of determining emissions to assess applicability, the Commission clarified that activities, emissions or equipment that are not required to be reported on annual TAC reports in Regulation Number 3, Part A, Section IX., or on ONGAEIR forms in Regulation Number 7, Part B, Section V. do not need to be considered when in stationary source-wide emission estimates for formaldehyde or hydrogen sulfide. Following existing exemptions in these programs streamlines requirements and creates consistency for both stationary sources subject to the rule and Division staff reviewing applicability.

Existing stationary sources (determining applicability with Sections III.D.1.a. for formaldehyde or III.E.1.a. for hydrogen sulfide)

Stationary source emissions-based applicability will be determined based on the highest reported emissions across the three years of TAC reports (calendar years 2023, 2024, or 2025), if available, or data reported to the Division on APENs. The Commission acknowledges that some stationary sources with wide variations in operating loads may opt to base their applicability on a 3-year average for calendar year 2023, 2024, and 2025 TAC reports. Since these reports involve historical reported data, the Commission also adopted an option for stationary sources to provide a one-time emission update report for their annual emissions of the relevant PTAC by January 1, 2027, based on their 2025 calendar year emissions (or the updated 3-year average if a stationary source chooses). This update can explain any alignments with the equipment and activity exemptions and the reporting thresholds for formaldehyde and hydrogen sulfide established in Regulation Number 3, Part A, Section IX.A.2. or Regulation Number 7, Part B, Section V. (for ONGAEIR sources) or any other explanation for the updates. For stationary sources who are not above the emission threshold, the Commission adopted a requirement that the stationary source maintain records of their emissions, including whether they based their emissions on the single highest year, a 3-year average, or an emission update report, and make the report available to the Division upon request.

The Commission based the thresholds for existing stationary sources on actual emissions because this was the basis of the data analysis used to identify the initial set of control strategy requirements and the potential universe of facilities that may require a non-categorical analysis. Because existing stationary sources are already built, prioritizing the control strategies at facilities that are actually emitting PTAC above a threshold will allow for the control strategies to be prioritized in order to align with legislative declarations aimed at meaningfully reducing exposures from PTAC from actual, not theoretical potential emissions. The Commission did not adopt on-ramp provisions for existing stationary sources who do not modify. Should a stationary source's emissions increase above the thresholds in a future year, such sources could become subject to PTAC emission control regulations at a future date, for example, at the next scheduled Commission review of the standards pursuant to Section 25-7-109.5(7)(d). This approach will allow time for the Division to assess what gaps may emerge based on yet-to-be-reported annual TAC emission reports, as well as other Division resource considerations. The Commission requests that the Division consider whether on-ramp provisions for existing stationary sources at the next review of the emission control regulations would be feasible or necessary.

New and modified stationary sources (determining applicability with Sections III.D.1.b. or III.D.1.c. for formaldehyde or III.E.1.b. or III.E.1.c. for hydrogen sulfide)

For new and modified stationary sources, the Commission based the thresholds on forecasted actual PTAC emissions. The Commission required a forecast period of five years, as this period aligns with the long-term review cycles of the emission control regulations and also the five-year renewal period for the Division's APEN program. The forecasted emission rates can take into account planned control device equipment or other operations reflecting projected business growth. While the Commission recognizes that some stationary sources may forecast emissions at or near full capacity, other stationary sources anticipated to be covered by the formaldehyde requirements,

particularly in the power generation sector, may have variable emissions in response to power demand loads or other predictable factors. Using forecasted actual PTAC emissions to assess applicability, similar to actual annual PTAC emissions, ensures strategic use of finite Division resources to implement the rule requirements while furthering legislative declarations aimed at meaningfully reducing exposures from PTAC emissions. A stationary source's potential to emit may be represented as its forecasted actual emissions. To reduce administrative burden, the Division will consider opportunities to utilize or integrate existing Division-approved processes to meet the requirement to submit forecasted actual emissions.

To address concerns with transparency and enforceability of these forecasted actual PTAC emissions, the Commission required forecasted emissions to be included at the time of permit application for the new or modified stationary source. If no permit application is required, the stationary source must report its forecasted emissions before the stationary source commences construction or reconstruction on any spark ignition stationary reciprocating internal combustion engine or stationary combustion turbine. Additionally, if a stationary source's forecasted actual emissions are below the threshold, the stationary source must track its post-operation actual annual emissions to ensure such emissions do not exceed the threshold for five years after commencing operation and maintain records. If actual emissions exceed the threshold, the stationary source must notify the Division and must comply with the rule, as applicable and by the prescribed deadline. If a stationary source anticipates operating its emitting equipment year-round, it would not be precluded from listing its full-capacity value as their forecasted actual PTAC emissions.

Location-based criteria

A secondary applicability criteria for existing and modified stationary sources of formaldehyde was added to further refine applicability by considering the location of the stationary source relative to locations with modeled cancer risks above a certain level and occupied areas that are in disproportionately impacted communities. The adopted emission control regulations do not include this additional location-based applicability criteria for the other PTACs because benzene, hexavalent chromium, and ethylene oxide do not have emissions thresholds and hydrogen sulfide does not have modeled cancer risk. Additionally, the Commission did not apply the secondary applicability criteria for new sources so as to provide an additional level of stringency in requirements applicable to new sources.

Due to the large number of formaldehyde-emitting sources, sources of formaldehyde are subject to the emission control strategies if they (1) actually emit (or are forecasted to emit) 4,000 pounds per year of formaldehyde, and are either (2) located within 1-mile of an occupied area that is within a disproportionately impacted community or (3) located within 1-mile from a census block that had a combined modeled cancer risk from carcinogenic PTAC of at least 20 excess cancer cases per one million people, based on modeled data in EPA's AirToxScreen 2020 with certain adjustments. The adjustments to AirToxScreen 2020 included the addition of toxicity factors used for each PTAC as promulgated by the Commission in Appendix B of Section II. of this Regulation

30, data adjustments established by the EPA in 2020 AirToxScreen Emissions Update Document to account for changes to modeling assumptions for the non-point oil and gas sector for the 2020 modeling year, and exclusion of any biogenic estimates for formaldehyde. The 2020 PTAC Cancer Risk Screening Area should not be interpreted as current modeled risk in any given area. The 2020 AirToxScreen provides a snapshot in time using emissions from calendar year 2020. Since 2020, emissions of PTAC could have changed, but this screening level tool reflects that latest version of modeled PTAC risk available for Colorado census blocks at the time of rule adoption.

The criteria of modeled cancer risk of at least 20 excess cancer cases was selected to prioritize which existing or modified stationary sources would be subject to these control regulations. This cancer risk level reflects the additive risk from four carcinogenic PTAC. The Division's staff research identified two other state programs where risk reduction action was required based on additive facility-wide cancer risk thresholds. Specifically, the Clean Air Oregon program has risk action levels of 25 in one million (voluntary risk reduction) and 50 in one million (toxics best available control technology) for existing sources, and the South Coast Air Quality Management District Rule 1402 has an action risk level of 25 in one million for control of toxic air contaminants from existing sources. The Commission recognizes that these other state programs have additive risk thresholds that are compared to the risk from a much larger list of carcinogenic pollutants, while the modeled risk adopted was limited to the four carcinogenic PTAC used in this analysis.

The PTAC cancer risk criterion identifies areas of the state with the highest modeled risk based on exposures. The Commission recognizes that modeled risk may change over time as EPA periodically releases new versions of AirToxScreen, which could change the areas that fall above the cancer risk threshold. The Division staff provided a guidance document that listed the census blocks that are above the cancer risk threshold as of the date of rule adoption and the Commission incorporated that guidance document into the rule. The incorporated census block list determined applicability of the cancer risk-based location criterion for this initial set of PTAC emission control regulations. As of adoption of the rule, Division staff published that list of census blocks and a geospatial information system shapefile on its website for sources to determine whether they are located within a 1-mile of these areas. The Commission also recommends that the Division revisit the list of census blocks at the next scheduled Commission hearing to review the PTAC emission control regulations and recommends that the Division create a mapping tool by July 31, 2026, to facilitate the identification of locations that meet this criterion.

The disproportionately impacted community criterion specifically addresses the statutory directive to prioritize reductions impacting disproportionately impacted communities in recognition that these areas may experience higher environmental and health burdens than other areas of the state. Combining proximity to disproportionately impacted communities, including mobile home parks, or larger areas defined at the census block group level with an option for sources to evaluate whether they are emitting PTAC near an occupied area, prioritizes reductions from sources that may expose human receptors to those PTAC emissions.

The Commission adopted a 1-mile buffer distance to be applied relative to the location of potentially applicable stationary sources emitting formaldehyde. If the facility is located in or within a 1-mile buffer area, or the source itself intersects with a 2020 PTAC Cancer Risk Area or an occupied disproportionately impacted community, the stationary source meets the relevant location-based applicability criterion. The buffer distance of 1-mile was adopted for both criteria for consistency and ease of review for both owner or operators of affected sources and Division staff assessing applicability. A 1-mile buffer distance is also used in Commission Regulation No. 3, Part B, Section III.B.5.d. and Regulation No. 27, Part B, Section II.A.6.

The Commission understands the Division will review ambient air toxics monitoring data when the Division reviews the identification of disproportionately impacted communities following the publication of updates to the Disproportionately Impacted Community Mapping tool, to evaluate whether additional communities need to be prioritized in the applicability location-criteria.

Emission Control Regulations by PTAC and Category

Benzene

Applicability: The Commission adopted a control requirement for existing and new petroleum refineries.

Exemptions: There are no exemptions from the benzene control requirement.

Rule Requirement: The Commission adopted more stringent leak detection and repair (LDAR) standards for components at petroleum refineries in Regulation Number 24.

Other Facilities with Benzene Emissions: As noted below, the Division's analysis identified additional facilities with emissions above a 4,000 lb/yr benzene emissions threshold that were not petroleum refineries and therefore not subject to the categorical LDAR requirement. During the rulemaking process, however, those sources provided additional information to Division staff demonstrating that they were either below the threshold, or that they had already completed a substantive analysis of feasible control strategies. The Commission, therefore, did not adopt requirements for these sources to complete a non-categorical analysis report.

The Commission did not adopt control requirements or non-categorical benzene analysis requirements for the upstream and midstream oil and gas segments as part of this rulemaking, due to anticipated co-benefit reductions from other regulations applicable to these segments and because improved TAC emissions information for these segments is expected to come from the revised annual reporting requirements in Regulation Numbers 3 and 7. The Commission asked the Division to provide updated information on benzene emissions and ambient air quality trends and report to the Commission on the possible need for PTAC emission control regulations, including from upstream and midstream oil and gas segments, no later than August 2028.

This is the first emission control regulation for PTACs completed by the Commission under HB22-1244. The Commission intends that the calculated cost figures for benzene in this rulemaking, while appropriate for this action, are not intended to be precedent setting for future rulemakings.

Chromium Compounds, Hexavalent (Hexavalent Chromium)

Applicability: The Commission did not adopt a de minimis emissions threshold for sources that have chromium electroplating or anodizing processes. The adopted requirements are based on the federal NESHAP for Hard and Decorative Chromium Electroplating and Chromium Anodizing Tanks (40 C.F.R. Part 63, Subpart N) and California Air Resources Board (CARB)'s Airborne Toxic Control Measure (ATCM) for Chromium Electroplating and Chromic Acid Anodizing Operations (division 3, chapter 1, subchapter 7.5, Section 93102 of Title 17 of the California Code of Regulations) rule. Neither of these rules have established minimum emission thresholds from chromium electroplating and anodizing tanks and fugitive emission sources that need to comply with their respective requirements.

Emission Limitations and Work Practices: While the federal NESHAP Subpart N requires control of emissions from decorative and functional plating and anodizing tanks, the Commission adopted additional requirements to reduce fugitive emissions and further refine the control requirements from these tanks. These electrolytic processes cause mists containing hexavalent chromium to be ejected from the plating tanks, which can eventually be emitted into outdoor air through building openings and vents. Other examples of fugitive emissions include dust containing hexavalent chromium emitted from compressed air operations, buffing and grinding operations, and cleaning. These and other types of fugitive emissions can contribute to ambient concentrations of hexavalent chromium near the source. Since many of the facilities are located in disproportionately impacted communities or are in close proximity to sensitive receptors, the Commission determined the additional requirements are necessary to reduce emissions of and exposure to this hazardous chemical in order to protect the health of Coloradans.

The Commission adopted an approved fume suppressant list, and their associated surface tension values and approved uses, that align with those that CARB adopted in 2023, and language that will allow the use of any other suppressant that has since been approved by CARB under their alternative fume suppressant procedures (division 3, chapter 1, subchapter 7.5, Section 93102.8 of Title 17 of the California Code of Regulations). The Commission finds that the list of approved suppressants ensures effective fume suppressants are used for the appropriate type of plating or anodizing operation and that lower, surface tension values are adhered to which are more stringent than the surface tension values in the federal NESHAP, to further reduce emissions.

The Commission adopted a requirement for the installation of building enclosures at sources of functional and decorative plating operations, and buffing, grinding, and polishing operations at such sources that use hexavalent chromium. The inclusion of

specific requirements for enclosure openings, such as those on opposite ends of the building or in proximity to sensitive receptors, which the Commission finds are necessary to protect public health from the emissions of hexavalent chromium. The Commission is allowing a two-year lead time for facilities to install these enclosures, which is consistent with the compliance schedule set by CARB in the ATCM.

The Commission is adopting a prohibition, effective after June 14, 2026, on the installation of new equipment, or modified equipment that involves the use or increase in use of hexavalent chromium for decorative applications. This action is based on the finding that safer, less toxic, and economically feasible alternatives are available, thus halting the creation of new sources of hexavalent chromium from this source type.

Performance Testing: The Commission adopted a requirement for all sources that continue to use hexavalent chromium in plating and anodizing operations to submit a performance test by April 15, 2027, according to procedures outlined in the federal NESHAP (40 CFR 63.347). The testing deadline allows for four months after the required deadline for sources to begin using an approved fume suppressant. The Commission finds that because the federal NESHAP Subpart N only required initial testing, the tests are outdated and often do not reflect current source operations, nor do they reflect change to the resulting emissions that may occur as a result of these more stringent surface tension requirements and approved suppressants.

Reporting and Recordkeeping: The Commission adopted a requirement for all sources affected by this regulation to submit an annual compliance report for any facility operating in the prior calendar year and to be due June 30, 2028, and annually thereafter. An annual compliance report is necessary to improve Division oversight, especially since many sources are not required to obtain a permit, and the federal NESHAP Subpart N does not require reporting for smaller sources. The Commission finds that the requirements and certifications will meet the goal of improving oversight from these source types.

Ethylene Oxide

Applicability: The Commission has adopted control regulations for sterilization facilities and finds that all three sterilizer facilities in Colorado will reduce emissions as a result of these emission control regulations. Further, any new sterilization facilities constructed in the state would also be subject to these controls, including the more stringent emission control requirements for new sources under 40 CFR Part 63, Subpart O (April 4, 2024). The emission control regulation does not cover sterilization facilities used for research and development or in hospitals, dentist offices, or veterinary clinics. While the Commission is not aware of any other types of sterilization facilities in Colorado that use ethylene oxide, the Commission understands that expanded reporting requirements for ethylene oxide will begin with reports due between June 2027 and June 2028 and it will evaluate whether additional types of sterilization facilities or other sources of ethylene oxide may present an opportunity for control strategies at its next review of the standards.

Emission Limitations: The Commission finds that the requirements in the federal NESHAP for sterilization facilities (40 CFR Part 63, Subpart O (April 4, 2024)) are well researched, health-protective, and generally effective at reducing ethylene oxide emissions in Colorado while accommodating a phased compliance schedule that recognizes sensitivities in the medical supply chain. To minimize potential conflicts with federal requirements, the Commission adopted revisions to Regulation 30 to include emission control regulations that largely incorporate by reference the federal standard that was finalized in April 2024. This incorporation ensures continuity even as the EPA undertakes a reconsideration of the federal rule, providing regulatory stability ahead of the State's April 2026 deadline for adopting PTAC emission control regulations. The Commission also made two revisions to the federal regulation to address Colorado-specific needs.

First, the Commission adopted technically feasible emission control requirements for one source, LivaNova (AIRS ID 059-1258), that was granted a presidential exemption under Section 112 of the Clean Air Act, 42 U.S.C. § 7412(i)(4) and 40 C.F.R. Part 63, Subpart A at § 63(j). The presidential exemption extends deadlines for compliance with the 2024 NESHAP Subpart O by two years. The Division understands that LivaNova is the only source that has been granted a presidential exemption at the time of this rulemaking. The Commission adopted standards for this source, beginning April 6, 2028, to require the facility to utilize a continuous emission monitoring system (CEMS) to demonstrate that the catalytic oxidizer controlling emissions from the sterilization chamber vents (SCV), aeration room vents (ARV), and chamber exhaust vents (CEV) complies with the applicable standards under 40 C.F.R. Part 63, Subpart O (April 5, 2024) § 63.362(b), (c), (d), (e), or § 63.362(i), with periodic reports to verify compliance. As an alternative to emission limitations adopted for sterilizers more generally, the Commission has adopted an operational standard to allow the source an option to limit annual ethylene oxide usage to 12 tons per year until the source can fully comply with 40 CFR Part 63, Subpart O (April 4, 2024), as incorporated into Regulation 30.

While the Commission accommodated LivaNova's request for additional time to comply with the CEMS requirements in Regulation 30, these accommodations were based only on the Commission's finding that meeting the requirements earlier than April 8, 2028, would be technically infeasible based on information LivaNova submitted. The Commission does not concede that the presidential exemptions preempt State authority under Section 112(d)(7) of the Clean Air Act, 42 U.S.C. § 7412(d)(7) to adopt more stringent emission control measures than the federal government requires. The Commission has clear and explicit authority under State law to adopt emission control regulations for PTAC, and nothing in CAA Section 112, including the presidential exemption authority, diminishes that authority. Additional extensions to the presidential exemption, if granted, will have no effect on compliance dates under the rule. Additionally, the Commission clarifies that April 8, 2028, is the date for demonstration of compliance for the CEMS at the LivaNova facility. This is the date by which full compliance with emissions standards must be demonstrated.

Exemptions: For small sterilizers that use less than 400 lbs annually (in any given 12 month period) of ethylene oxide, the Commission exempted these sources from the

State PTAC emission control regulation. This exemption is anticipated to streamline compliance demonstration provisions for these small sources.

Formaldehyde

The Commission adopted revisions in Regulation 30 to establish emission control strategies for new and existing stationary sources that emit formaldehyde. Regulation 30 revisions also include the non-categorical requirements for site-specific analysis for applicable formaldehyde sources.

Applicability: The Commission adopted applicability criteria for existing stationary sources (those that commenced construction before September 1, 2026) that have actual emissions of formaldehyde at or above 4,000 pounds per year and meet the location-based criteria described above. For modified sources (those that modify after September 1, 2026), the applicability criteria is forecasted actual formaldehyde emissions and location-based criteria. For new sources, (those that commenced construction on or after September 1, 2026), the applicability criteria is forecasted actual formaldehyde emissions at or above 4,000 pounds of formaldehyde per year.

The Commission adopted the Division's proposed compliance timelines for sources that had not yet commenced construction as of September 1, 2026, and for sources that had commenced construction but not operation as of September 1, 2026. However, the Commission wanted to give more time for compliance to sources that had commenced construction and operation as of September 1, 2026. For this latter group, the compliance deadline was extended from September 1, 2026, to January 1, 2027.

Exemptions: The Commission exempted from the applicability determination equipment or activities that are not required to be reported on TAC reports under Regulation Number 3, Part A, Section IX.A.2. or Regulation Number 7, Part B, Section V.

Emission Limitations: The Commission adopted emission control requirements for stationary source engines and turbines in Regulation Number 30.

Engines

The Commission recognizes that several regulations adopted in recent years address emissions from existing spark ignition gas-fired engines in the upstream and midstream oil and gas segments. The nitrogen oxides (NOx) emissions limitations for engines adopted in Regulation Number 26 in 2020 and 2023 as well as the NOx intensity and use practices adopted revised in Regulation Number 7 in 2023 and further expanded in 2025 achieve formaldehyde emissions as a co-benefit to reducing NOx emissions from various combustion equipment in oil and gas operations. Additionally, revisions to Regulation Number 7 adopted in 2021 and 2024 that reduce greenhouse gas emissions from fuel combustion equipment in the midstream segment are also expected to achieve reductions in formaldehyde. The recent adoption of these NOx and greenhouse gas reduction measures informed the Commission's decision to not adopt extensive, additional categorical requirements for existing engines and turbines located at sources

that met the applicability criteria for formaldehyde emissions in the upstream and midstream oil and gas segments.

However, to ensure formaldehyde emission reductions and to complement the federal NESHAP, Subpart ZZZZ (August 30, 2024), including where it defers to the federal NSPS, Subpart JJJJ (August 30, 2024), the Commission adopted Subpart ZZZZ requires for existing spark ignition (SI) reciprocating internal combustion engines (RICE) (constructed or reconstructed before September 1, 2026) and additional emission limitations for new SI RICE (constructed or reconstructed on or after September 1, 2026) at new and existing applicable sources.

For new or reconstructed non-emergency SI RICE rated at or above one hundred (100) horsepower (hp), the Commission adopted a requirement for the engines to comply with the carbon monoxide (CO) emissions limit of 0.40 g/hp-hr, unless otherwise specified in Table 1 at Section III.D.2. In many cases, this CO limit is more stringent than Subpart ZZZZ requirements and was identified to be a state-of-the-art, technically feasible level of control in 2023 by the New Jersey Department of Environmental Protection. The Commission understands that CO is commonly used as a surrogate emission limit and pollutant to demonstrate formaldehyde emission reductions. However, rich burn engines above 500 horsepower at major sources of HAP will be subject to formaldehyde specific requirements under Subpart ZZZZ and, therefore, the Commission exempts these engines from the CO limits in Table 1. This exemption will also avoid a conflict with the Subpart ZZZZ requirements. The Commission also adopted alternative compliance options, based on percent reductions of CO, that align with New Jersey state-of-the-art program for two-stroke lean burn (LB), four-stroke LB, and rich burn (RB) SI RICE. The adopted emission limitations do not apply during periods of startup or shutdown but owners or operators must maintain and operate the engine in a manner consistent with good air pollution control practices to minimize emissions during these periods.

The Commission also adopted annual inspection and adjustment (i.e., tuning) requirements for engines. Owners or operators must maintain records of compliance with the state-only annual tuning requirements or federal tune-up or adjustment requirements, if applicable. Owners or operators must also conduct applicable performance tests and keep records and submit reports in alignment with federal NESHAP or NSPS provisions, with the expectation that the testing and associated records will demonstrate compliance with the more stringent CO emission limitations.

The Commission adopted several exemptions from the engines emissions control strategies. Specifically, the Commission exempted non-road engines, emergency power generators, and engines being removed or electrified to meet requirements of the midstream greenhouse gas reduction program. These exemptions are similar to exemptions the Commission has previously adopted in its engine rules. The Commission also adopted an exemption for equipment and activities also exempt from the annual emissions reporting requirements in Regulation Number 3, Part A, Section IX. or Regulation Number 7, Part B, Section V.

Turbines

To ensure formaldehyde emission reductions and to complement the federal NESHAP, Subpart YYYY (March 9, 2022), the Commission adopted tuning requirements for existing stationary combustion turbines (constructed or reconstructed before September 1, 2026) and additional emissions limitations for new stationary combustion turbines (constructed or reconstructed on or after September 1, 2026) at new and existing applicable sources.

For new or reconstructed stationary combustion turbines, the Commission adopted a requirement for the turbine to comply with the CO emissions limits in Table 2 in Section III.D.3., unless a turbine is subject to Subpart YYYY (i.e., constructed or reconstructed on or after January 14, 2003, and located at a major HAP source). The Commission also adopted an option to optionally comply with the Subpart YYYY formaldehyde emissions limitations instead of the CO emissions limits. Similarly to the engine requirements described above, the Commission adopted these CO emissions limits as a surrogate to demonstrate formaldehyde emissions reductions. Turbines subject to Subpart YYYY must already comply with the formaldehyde requirements in that subpart. However, the CO limits, or Subpart YYYY compliance option, adopted by the Commission apply more broadly than Subpart YYYY in that they apply to new turbines at area HAP sources. These CO limits were also identified to be a state-of-the-art, technically feasible level of control by the New Jersey Department of Environmental Protection. In addition, all turbines must be operated and maintained based on manufacturing specifications or equivalent to the extent practicable, and consistent with technological limitations and good engineering and maintenance practices.

For a turbine optionally complying with the Subpart YYYY, a turbine may comply with the formaldehyde emission limitation with or without an oxidation catalyst. If complying without an oxidation catalyst, the owner or operator must develop and comply with an operating parameter monitoring plan that ensures the unit demonstrates compliance with the applicable emissions limitation. The turbine owner or operator will determine the appropriate parameters and the operating ranges for its plan, and may utilize information in performance tests or in manufacturer recommendations, as applicable. The Division has authority to review and revise plans upon request. This approach is similar to the federal Subpart YYYY framework and ensures that stationary combustion turbines have flexibility to determine how to ensure continuous compliance with the formaldehyde emission limits under Division oversight.

The Commission also adopted annual inspection and adjustment (i.e., tuning) requirements for turbines. Owners or operators must maintain records of compliance with the tuning requirements. Owners or operators must also conduct monitoring, performance tests, and keep records and submit reports in alignment with federal NESHAP provisions, with the expectation that the testing and associated records will demonstrate compliance with the more stringent CO emission limitations.

Similar to the adopted engine requirements, the Commission adopted an exemption for equipment and activities also exempt from the annual emissions reporting requirements in Regulation Number 3, Part A, Section IX. or Regulation Number 7, Part B, Section V.

Other Facilities with Formaldehyde Emissions

As noted below, the Division's analysis identified additional facilities with emissions above the 4,000 lb/yr formaldehyde threshold that were not subject to a formaldehyde categorical requirement. During the rulemaking process, several of these sources provided additional information to Division staff demonstrating their facility's emissions were either below the threshold, or the facility had already completed substantive analysis of feasible control strategies. The Commission, therefore, removed these sources from the requirement to complete a non-categorical analysis report. The remaining sources, named in Section III.F., will either follow the non-categorical requirements to inform potential future control strategies, or will submit to staff an approved form and testing method to correct formaldehyde emissions, as applicable.

Hydrogen Sulfide

Applicability: The Commission adopted emission control regulations for two categories of stationary sources that Division staff research determined were above the emission threshold of 10,000 pounds per year. The two categorical emission control strategies impact asphalt and roofing manufacturing operations and manure digesters and will impact three sources in the state. The Commission adopted applicability criteria for existing stationary sources (those that commenced construction before June 14, 2026) that have actual emissions of hydrogen sulfide at or above 10,000 pounds per year. The emission control strategies will also apply to any new or modified facilities with these types of operations above the emission threshold. For modified sources (those that commenced construction before June 14, 2026, and modify after June 14, 2026), the applicability criteria is forecasted actual hydrogen sulfide emissions. For new sources (those that commenced construction on or after June 14, 2026), the applicability criteria is forecasted actual hydrogen sulfide emissions at or above 10,000 pounds of hydrogen sulfide per year.

Emission Limitations:

The Commission adopted emission control requirements for two different categories of emissions that were above the applicability threshold.

Asphalt and Roofing Manufacturing:

The Commission adopted emission control regulations for equipment and activities with hydrogen sulfide emissions at all new, modified, or existing asphalt or roofing manufacturing operations to install technically feasible control technology abatement devices such as regenerative thermal oxidizers (RTOs). These sources must also demonstrate a measurable reduction in hydrogen sulfide and benzene emissions with an applicable destruction efficiency such that emissions from asphalt storage, surge, and mixing tanks; sealant kettles; coating and applicator equipment; and loadout racks are controlled. The sources must perform periodic performance testing for VOC or H₂S, demonstrating an appropriate destruction efficiency from the abatement devices. Sources must also conduct manufacture-specified inspection and maintenance of

abatement devices; perform weekly visual inspections and maintenance of asphalt processing or roofing manufacturing equipment, as applicable; maintain regular inspection records, maintenance records, and cleaning records of asphalt storage and surge tanks; and replace filters to minimize fugitive emissions from asphalt manufacturing processes. Many of these requirements are aligned with the latest Division-approved operations and maintenance plan to preserve site-specific flexibility, when warranted.

Exemptions: The Commission exempted from the applicability determination equipment or activities that are not required to be reported on TAC reports under Regulation Number 3, Part A, Section IX.A.2. or Regulation Number 7, Part B, Section V. The Commission adopted a conditional exemption for asphalt processing and roofing plants that are actively seeking to limit their H₂S emissions through a permit modification. At the time of rule adoption, there were two sources with pending administratively complete permit applications for permits that will allow for control equipment to be installed to reduce H₂S emissions below the emission threshold. This exemption is conditioned on cooperation with the Division to obtain the permit, timely construction occurring after the permit is issued, and ensuring actual H₂S emissions remain below the emission threshold after the permitted control begins operation.

Manure Digester Facilities:

The Commission adopted provisions for all new, modified, or existing anaerobic digester facilities that use manure as a feedstock to minimize hydrogen sulfide emissions including fugitive gas releases. The emission control regulations require these sources to continuously inject non-corrosive, hydrogen sulfide reducing additive directly into the liquid slurry of the manure receiving pit to chemically bind H₂S, precipitate sulfides, and reduce H₂S formation by at least 500 ppm or lower; maintain existing covers on the tanks and/or lagoons associated with anaerobic digestion process; regularly remove manure from the manure receiving pit to limit H₂S formation; and interconnect the vapor head space of manure storage tanks with the anaerobic digesters to effectively capture H₂S while maintaining negative pressures and preventing H₂S corrosion and build-up of volatile gas mixtures especially during agitation. The sources must demonstrate additional best management practices which include installing and operating a gas-flow metering and analysis using a Division-approved system to continuously measure H₂S concentrations from the anaerobic digester and scrubber outlets; maintain flares or enclosed combustion systems to safely oxidize excess H₂S when it is not routed to energy recovery or upgrading systems; install and maintain approved filtration equipment to remove H₂S, CO₂, and other contaminants prior to pipeline injection; and perform continuous monitoring of fugitive H₂S with periodic calibration and maintenance to keep H₂S within safe levels, ensuring scrubber and filter efficiency.

Non-Categorical PTAC emission sources

The Commission adopted a provision applicable to two sources that meet certain applicability criteria for formaldehyde where the primary formaldehyde emissions are not from a spark ignition engine or combustion turbine. One source is required to submit

additional information to the Division about its source-specific emissions and opportunities for technically feasible and cost-effective emission reductions. The other source is only required to submit source-specific emissions information. The required analysis regarding feasible control measures must include actual annual emissions of applicable PTAC from activities or equipment and an analysis of what control strategies are technically feasible and cost-effective to reduce those emissions. Sources must provide information for each emission point or activities that emits formaldehyde.

To supplement guidance already included in the existing guidance such as “Overview of Reasonably Available Control Technology Analyses for Major Sources in the Denver Metro and Northern Front Range (DMNFR) 2008 Ozone Nonattainment Area” guidance from May 2017, the Division will engage with the subject source and provide additional direction to support the development of site-specific PTAC analysis. The subject source is not required to implement any of the PTAC emission reduction measures identified in the site-specific analysis, absent a separate Commission rulemaking to adopt such PTAC emission reduction measures.

The Commission also adopted an option for the subject source to submit documentation to the Division to demonstrate that its emissions are below the applicability threshold or outside of the covered location criteria to avoid the site-specific emissions and controls analysis requirement.

New and modified sources

The Commission did not, at this time, adopt a broadly applicable emissions control strategy for new and modified sources that were not specifically identified as applicable non-categorical sources from the Division’s data analysis. A potential framework for permitting new and modified sources of PTAC has been presented to the General Assembly for consideration. The framework includes steps to assess project or facility risk from PTAC emissions. Depending on the risks estimated, the framework requires various actions to be taken by the source. One action is referred to as “TBACT” that would require a facility-specific analysis of feasible controls—similar to the above-described requirement for existing sources—to be completed if certain criteria are met.

Because this framework is being evaluated by the General Assembly, the Commission did not adopt requirements for new or modified sources that are not subject to category-specific requirements to submit PTAC emissions and control analyses at this time. The Commission determined that it was appropriate to wait for additional direction from the General Assembly regarding a potential PTAC permitting program to avoid conflict or duplication with that potential program.

Further, these revisions will correct any typographical, grammatical, and formatting errors found within the regulation.

Incorporation by Reference

The Commission will update regulatory references as needed as opportunities arrive.

Additional Considerations

There are no direct federal requirements that correspond with the new rule. However, to the extent the new rule could be construed to exceed or differ from any related requirements of federal law, the Commission provides the following additional statement, consistent with § 25-7-110.5(5)(a), C.R.S.:

(I) Any federal requirements that are applicable to this situation with a commentary on those requirements;

Emission control strategies for identified PTAC address equipment and operations in several sectors that are also subject to federal NESHAP/MACT or NSPS requirements. Discussed further under each PTAC below, Regulation Number 30 revisions apply on a broader basis, extend requirements to sources not subject to federal regulations, or complement the federal regulations with additional state-only requirements.

Benzene: The Commission adopted revisions in Regulation Number 24, Part B, Section VI. to modify the definitions of vapor loss and leak at petroleum refineries and modify monitoring programs to require recording the recording and, subsequently, repair of leaks. These facilities may also be subject to EPA's NSPS J, NSPS Ja, NSPS XX, NSPS GGG, NSPS GGGa, NSPS QQQ, NESHAP CC, NESHAP R, and NESHAP UUU.

The Commission previously incorporated by reference several federal NSPS and NESHAP in Regulation Numbers 6 and 8 that address benzene emissions from a variety of facilities including, but not limited to, petroleum storage and oil and gas operations; 40 CFR 60, Subparts Kb, VV and VVa, GGGa, OOOO and portions of OOOOa and OOOOb; 40 CFR 63, Subparts G, CC, YY, HHH, FFF, and DDDD. 40 CFR 63 Subpart ZZZZ is also expected to reduce benzene emissions, which the Commission incorporates by reference with this rulemaking action for applicable sources. 40 CFR 63, Subparts BBBB and CCCCC address benzene emissions from gasoline distribution at major and area sources, which apply at the federal level. The Commission has adopted provisions in Regulation 24 that align with many of the federal requirements in Subparts BBBB and CCCCC. The Commission adopted updated definitions, processes and subsequent monitoring, recording and repair of leaks to address benzene emissions at petroleum refineries. The refinery is currently subject to LDAR requirements in NSPS GGG and VV. However, the proposed hydrocarbon detection limits are more stringent than the ones in the federal rules and do not conflict with the federal requirements.

Chromium Compounds, Hexavalent: The federal NESHAP at 40 CFR 63, Subpart N for chromium electroplating and anodizing tanks controls hexavalent chromium emissions from many sources applicable to the adopted emission control regulations for hexavalent chromium. Subpart N contains provisions for both major and area sources performing either decorative or functional

electroplating and anodizing. Requirements include use of fume suppressants, composite mesh-pad systems, or packed-bed scrubbers and establishes a total chromium emission limitation. The adopted provisions in Part B, Section III. of this Regulation 30 create state-only requirements to address additional sources of hexavalent chromium emissions in the hard and decorative chromium plating and anodizing processes for area or major sources of HAP that meet applicability criteria.

Ethylene Oxide: The federal NESHAP at 40 CFR Part 63, Subpart O requires emission control from all of the sterilizers in Colorado as well as any new sterilizer being built. In addition, federal FIFRA rules (Interim Decision authored on January 7, 2025, Case No. 2275 EPA-HQ-OPP-2013-0244) require monitoring of room air emissions inside these facilities. The Commission incorporated Subpart O by reference in this rulemaking action. This rule strengthened federal regulation of ethylene oxide emissions from new and existing sterilization facilities through a comprehensive set of requirements aimed at more stringent emission limits for various equipment and fugitive emission sources. The Commission also adopted state-only emission control regulations that are more stringent than federal requirements for specific sources.

Formaldehyde: The federal NESHAP at 40 CFR 63, Subpart ZZZZ (August 30, 2024) targets HAP reductions, with a focus on formaldehyde limits, or alternative limits on carbon monoxide, by establishing emission and operating limits, inspection and maintenance actions, and testing requirements that apply to different engine categories. For most newer engines, an alternative set of compliance demonstrations is available by pointing to the requirements for new or reconstructed spark ignition (SI) reciprocating internal combustion engines in the NSPS, 40 CFR 60, Subpart JJJJ for emission limitations and compliance requirements. In deferring to Subpart JJJJ, sources that demonstrate compliance with Subpart ZZZZ through the NO_x, VOC, and CO emission limits, CO is a federally recognized surrogate for formaldehyde and is therefore the most relevant standard for demonstrating formaldehyde reductions.

The federal NESHAP at 40 CFR 63, Subpart YYYY (March 9, 2022) sets formaldehyde emission limits for stationary combustion turbines that commenced construction or reconstruction on or after January 14, 2023, and are located at major HAP sources. The primary formaldehyde numeric limit established in this subpart is 91 ppbvd or less corrected to 15% O₂ (dry basis), except during turbine startup. Subpart YYYY also establishes operating limitations that require maintenance of a rolling average catalyst inlet temperature for turbines that comply by using an oxidation catalyst. Compliance with the Subpart YYYY requirements may be demonstrated through performance testing using EPA Method 320/ASTM D6348-12e1 or an EPA approved alternative, and through ongoing monitoring scheme. There are also federal NSPS for stationary gas turbines at 40 CFR Part 60, Subpart GG, KKKK, and KKKKa, which are incorporated into the Commission's Regulation Number 6, except for NSPS Subpart KKKKa, which was finalized by EPA in January 2026. The NSPS

establishes emission limits for NO_x and SO₂ and associated testing, monitoring, and reporting requirements but does not include emission limits for carbon monoxide.

In this rulemaking action, the Commission incorporated provisions of both 40 CFR 63 Parts YYYY and ZZZZ into Regulation Number 30, in some parts, through incorporation by reference. Major sources of HAP with applicable engines or turbines are already subject to federal requirements. New or reconstructed (after June 12, 2006) engines at area sources must comply with Subpart ZZZZ by instead complying with NSPS JJJJ, pursuant to 63.6590(c). Additionally, the Commission adopted state-only requirements for new smaller SI RICE at both area and major sources of HAP that are more stringent than federal ZZZZ formaldehyde (or surrogate CO) limits. For turbines, the CO limits adopted in the rule only apply to turbines that commenced construction or reconstruction on or after September 1, 2026, that are located at area sources of HAP.

Hydrogen Sulfide: Hydrogen sulfide is not designated as a HAP under the CAA, Section 112, so there are no federal NESHAP/MACT that directly control H₂S. There are some NSPS that indirectly control H₂S as a result of required reductions in VOC or SO₂ emissions. For the asphalt and roofing material manufacturing plants that are installing RTOs, the requirement adopted by the Commission in Regulation Number 30 reduces H₂S but also relates to Colorado's obligations to address ozone precursor pollutants under the state implementation plan (SIP). There are no federal requirements that relate to the requirements adopted by the Commission to reduce fugitive emissions of H₂S at the manure anaerobic digester.

(II) Whether the applicable federal requirements are performance-based or technology-based and whether there is any flexibility in those requirements, and if not, why not;

The federal rules discussed in (I) are primarily technology-based in that they largely prescribe the use of specific technologies or work practices to comply. The proposed state-only requirements are technology- and performance- based and to allow for maximum flexibility for demonstrating compliance.

(III) Whether the applicable federal requirements specifically address the issues that are of concern to Colorado and whether data or information that would reasonably reflect Colorado's concern and situation was considered in the federal process that established the federal requirements;

The EPA develops NSPS or NESHAP considering national information and data, not Colorado specific issues or concerns.

Benzene: The above referenced federal requirements concerning petroleum storage are not as stringent as those adopted by the Commission in Regulation Number 30.

Chromium compounds, hexavalent: 40 CFR Part 63 Subpart N does not adequately cover fugitive releases of hexavalent chromium of concern to Colorado. The adopted provisions add requirements to address fugitive emissions and increase the stringency of requirements for tanks that are controlled by fume suppressants by requiring approved fume suppressants and more stringent surface tension requirements. Additionally, the adopted provisions require one-time testing requirements for tanks and prohibit new decorative chromium plating operations from using and emitting hexavalent chromium.

Ethylene Oxide. 40 CFR Part 63 Subpart O largely addresses Colorado's objective to reduce emissions from its sterilization facilities. However, as a result of concerns regarding potential continuous monitoring costs for small sterilizers, the rule includes an optional method for these sources to demonstrate compliance. Additionally, at the time of adoption, the EPA has announced reconsideration of Subpart O. Adopting state-only requirements in this Regulation 30 helps address a concern that changes at the federal level will make it more difficult to achieve emissions reductions for these facilities.

Additionally, regarding the one facility that has been granted a federal exemption to Subpart O, the Commission notes that Section 112(d)(7) of the Clean Air Act preserves the authority of States to adopt and enforce more stringent emission limitations or standards despite any presidential proclamations. That section reads, "No emission standard or other requirement promulgated under this section shall be interpreted, construed or applied to diminish or replace the requirements of a more stringent emission limitation or other applicable requirement established pursuant to section 7411 of this title, part C or D [of the Clean Air Act], or other authority of this chapter or a standard issued under State authority." These PTAC emission control regulations are issued under State authority.

Formaldehyde: NESHAP Subpart ZZZZ does not have as stringent formaldehyde limits, in the form of a CO surrogate, as other states for new smaller (≥ 157 bhp) SI RICE at area and major sources. NESHAP YYYY does not include turbines at area sources in their source category classification. The adopted state-only formaldehyde, or CO surrogate, emission limits for new SI rice and new stationary combustion will achieve reductions of formaldehyde.

While the NESHAP YYYY includes formaldehyde emission limits only for turbines at major sources of HAP, the Commission adopted applicability criteria based on lower thresholds than the federal HAP source designation in recognizing the need to address PTAC at lower levels than federal regulations due to pollutant toxicity and consideration of multiple exposure.

(IV) Whether the proposed requirement will improve the ability of the regulated community to comply in a more cost-effective way by clarifying confusing or potentially conflicting requirements (within or cross-media), increasing certainty, or preventing or reducing the need for costly retrofit to meet more stringent requirements later;

The adopted rules will provide certainty to sources, by providing specified technology-based performance standards and clear timing requirements. The adopted rules aid Colorado in achieving emissions control requirements as directed by Section 25-7-109.5(7) and provide regulated entities with flexibility to identify and cost-effectively employ available emissions control technologies to reduce PTAC emissions in the state.

(V) Whether there is a timing issue which might justify changing the time frame for implementation of federal requirements;

Ethylene Oxide: The federal NESHAP (40 CFR Part 63 Subpart O) allows for sources to petition for a presidential exemption. One source in Colorado was granted this exemption; however the basis of that exemption was not provided to the Division or Commission staff. Based on the Division's research, which included a site visit to this facility, this source can reasonably comply with the requirements in the federal rule, with the exception of group 2 room air emissions. The Division has provided an interim compliance mechanism for the group 2 room air emissions that is technically feasible.

These are state-specific rules that are not addressing federal requirements. Thus, there are no timing issues related to other applicable NESHAP or NSPS for the other four PTAC.

(VI) Whether the proposed requirement will assist in establishing and maintaining a reasonable margin for accommodation of uncertainty and future growth;

The adopted rules will assist in establishing and maintaining a reasonable margin for accommodation of uncertainty and future growth as it does not limit sources to specific technology to comply and aligns with regulations seen in federal and other state programs. Additionally, the rules establish reasonable times for meeting PTAC emissions control requirements. The rules afford the ability to affect required emissions controls through alternative compliance measures where appropriate and establish multiple pathways for compliance.

(VII) Whether the proposed requirement establishes or maintains reasonable equity in the requirements for various sources;

The rule effectively targets sources of PTAC in Colorado and focuses on achievable emissions reductions for these PTAC. Additionally, in adopting emission thresholds for applicability for certain pollutants, the Commission ensured the criteria for making sources subject to the emission control regulations were equitable.

(VIII) Whether others would face increased costs if a more stringent rule is not enacted;

If a more stringent rule is not enacted, public health and welfare and the environment could face increased costs associated with delays in implementation of appropriate PTAC measures.

(IX) Whether the proposed requirement includes procedural, reporting, or monitoring requirements that are different from applicable federal requirements and, if so, why and what the “compelling reason” is for different procedural, reporting, or monitoring requirements;

There are separate requirements to report or monitor PTAC from applicable federal regulations. Federal regulations have different monitoring and reporting thresholds for different types of source categories. In order for Colorado to protect public health and welfare and the environment, state-specific regulations that require more stringent or specific requirements are necessary. The rule addresses PTAC-specific considerations under Section 25-7-109.5 (1)(d), C.R.S., which states, “Emissions of toxic air contaminants should be monitored and controlled to levels that avoid, minimize, or mitigate harm to public health and the environment” as well as (2)(b), which requires the Commission to “Analyze public health risks from toxic air contaminant emissions from sources based on verified science and reporting of emissions data, including consideration of background levels in the ambient air.”

Where applicable, the Commission has not adopted additional procedural, reporting or monitoring requirements where the federal requirements satisfy Colorado specific concerns.

(X) Whether demonstrated technology is available to comply with the proposed requirement;

In aligning with requirements in Section 25-7-109.5(7)(a), C.R.S., the Commission adopted provisions that may require technology that has been demonstrated as available and technically feasible in other states or federal regulations.

(XI) Whether the proposed requirement will contribute to the prevention of pollution or address a potential problem and represent a more cost-effective environmental gain;

Further quantification of emissions reduced and related cost effectiveness of the regulations are outlined in the rulemaking materials such as the Economic Impact Analysis and various technical support documents

(XII) Whether an alternative rule, including a no-action alternative, would address the required standard.

The Commission is required to adopt emission control regulations that reduce emissions of each PTAC and prioritize reductions in disproportionately impacted communities with multiple sources of emissions of PTACs. A no-action alternative would not meet the statutory requirements. Alternative rules could

address this directive, such as different applicability criteria, compliance timelines, and emission control requirements. However, the Commission believes that for this initial promulgation of PTAC emission control strategies, the adopted applicability criteria for new and existing stationary sources as well as the adopted categorical and non-categorical requirements best meet the statutory considerations in Section 25-7-109.5(7)(b), C.R.S.

Findings of Fact

After considering all the information in the record, the Commission hereby makes the following determinations pursuant to section 25-7-110.8, C.R.S.:

- (I) These rules are based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.
- (II) Evidence in the record supports the finding that the rules shall result in a demonstrable reduction of harmful PTAC emissions and will enable the Commission to satisfy the requirements of section 25-7-105(1), C.R.S.
- (III) Evidence in the record supports the finding that the rules shall bring about reductions in risks to human health and the environment that justify the costs to implement and comply with the rules.
- (IV) The rule is the most cost-effective to achieve the necessary and desired results, provide the regulated community flexibility, and achieve the necessary reduction in air pollution.
- (V) The rule will maximize the air quality benefits of regulation in the most cost-effective manner.

Editor's Notes

History

New rule eff. 03/17/2025.

Entire rule eff. 11/14/2025.

Entire rule eff. 06/14/2026.